WELL WORKOVER PROGRAM

AZADEGAN OIL FIELD

WELL WORKOVER#1 PROGRAM

AZNS-075 (PAD #80), GADVAN WELL

Drilling Department

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45ºc in summer, and Highest Temperature is 52ºc. The Lowest Temperature is -4.4ºc in winter. Maximum Rainfall is 78 mm in February, Winter Season.

This Document covers the work over of well AZNS-075. Main headlines of program are de-complete the well and run new completion string.



Figure 1: AZNS Oilfield Location Map

1.2 Well Information

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	South Azadegan
Type Of Operation:	Workover
Well Name / Type :	AZNS-075 / Gadvan well in Block III
Pad Number:	80
Well Type:	Deviated Well
Wellhead Coordinate:	N: 3,436,066 E: 186,427
Gadvan E.P. Coordinate:	N: 3,436,066 E: 186,469
TD Coordinate:	N: 3,436,066 E: 186654.6
Well Objective:	De-complete the well, Side Track the Well & Re-complete the well,
RTE:	12.95 m
GLE:	3.95 m
Well Duration :	141.4 days (Main Plan) + 149.2 Days (with Contingency Plan)
Total Depth (MD):	3888.5 mMD
Producer Formation:	Gadvan (5" Liner)
Formation TD :	Gadvan
Reservoir :	Gadvan

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.
Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Block I is located on the north part of the CONTRACT AREA in north of well AZN-010. Gachsaran formation is characterized by abnormal high pressure in this block.
Block II is located between well AZN-010 and AZN-011. In this block, Gachsaran formation is characterized by pressure tending to normal
Block III is located on the south of line well AZN-011. In this block, Gachsaran formation is characterized by normal pressure and all drilled wells in this block suffered from complete loss at Pabdeh formation.

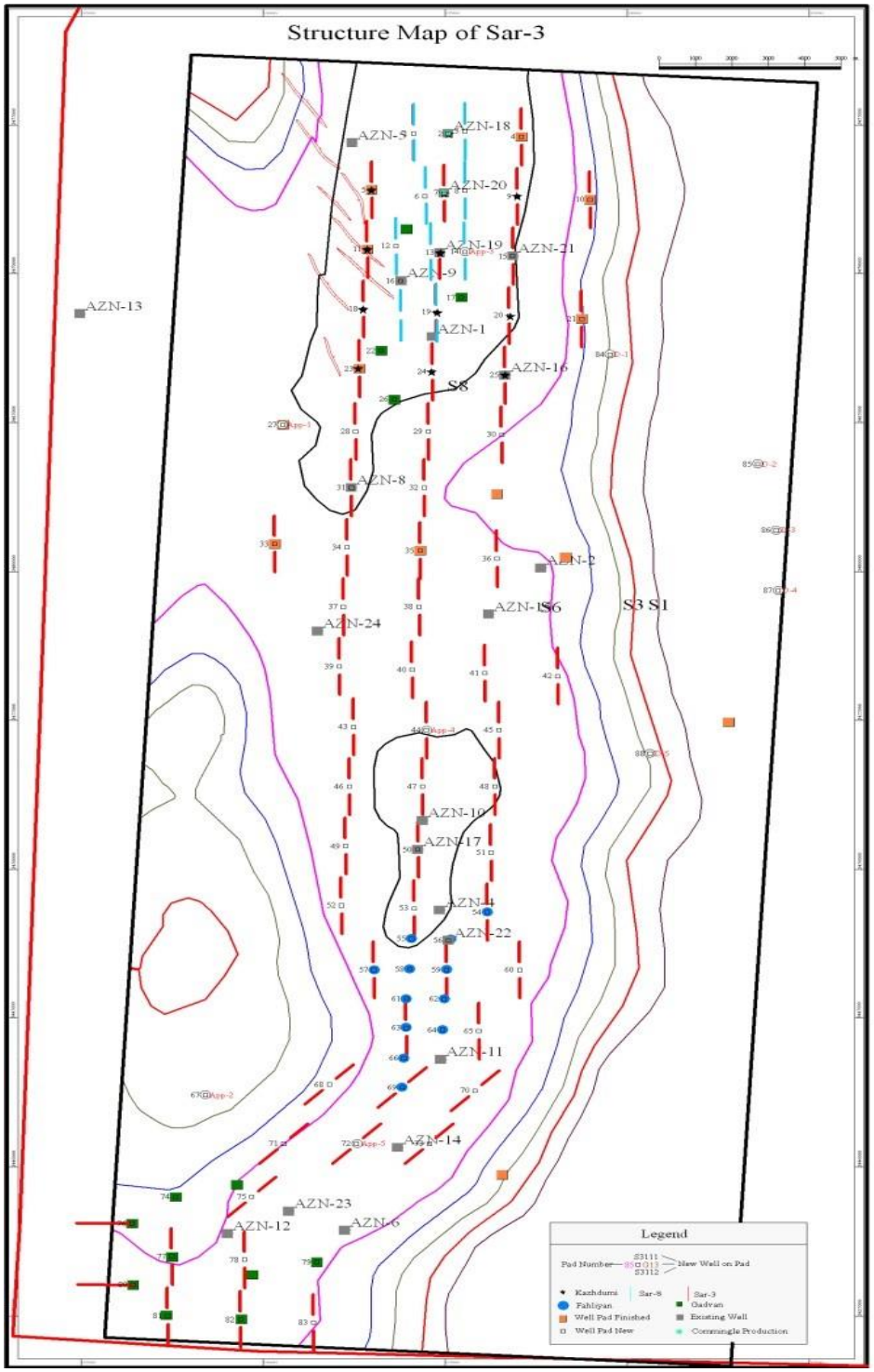


Figure 2: AZNS Area Distribution

1.4 Well Pad Specification

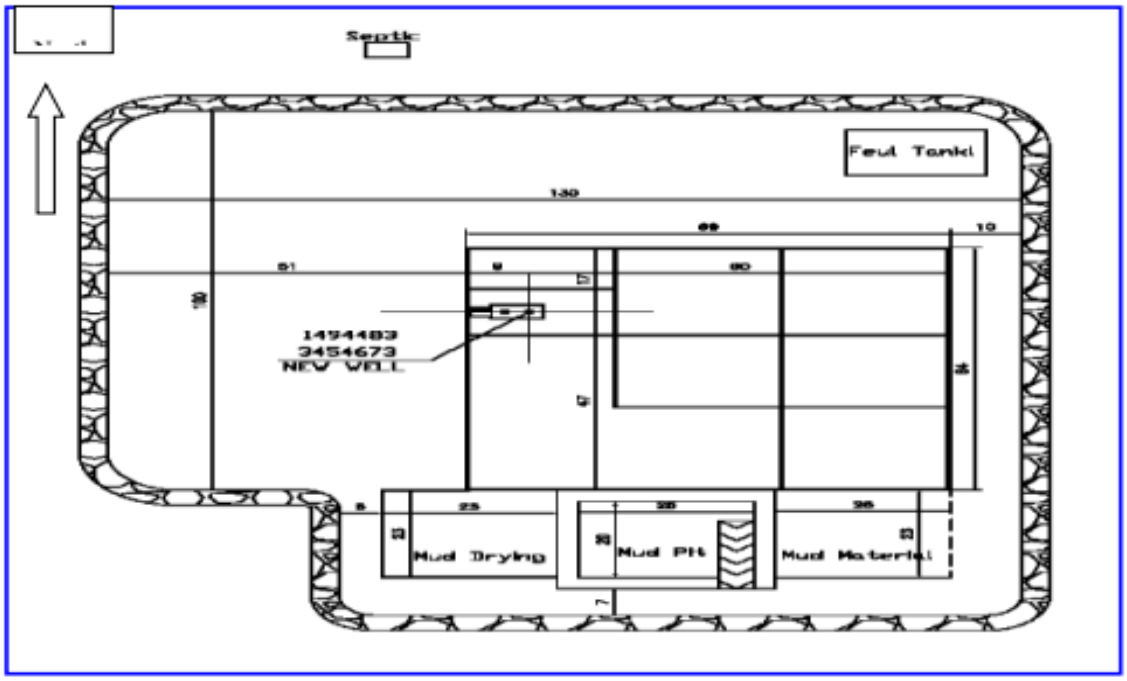
Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

Compressive Strength for reinforced concrete after 28 days: 280 kg/cm2

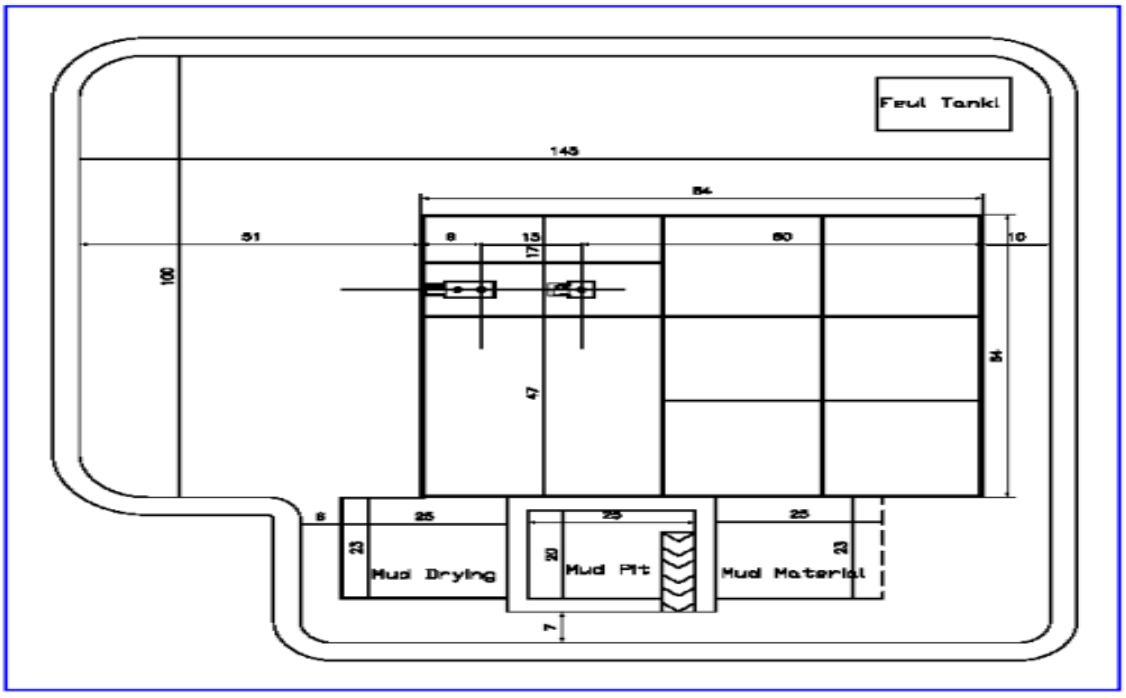
- Fuel tank concrete pond dimension: Length 20m×Width 12m×Height 0.5m
- Drainage: Width 30cm × Depth 30cm
- Burning pit: 15m×15m square, located on South-East or South-West with at least 350m distance from well head
- Cellar to cellar distance in multi-well well pad: 15m
- Conductor pipe depth under cellar bottom is 5.5m. The conductor pipe will be driven while constructing the well pad.
- Mud material dimension is 25m*35m +granular surface
- Mud drying dimension is 25*35m + concrete surface
- Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

1. Types of well pads:

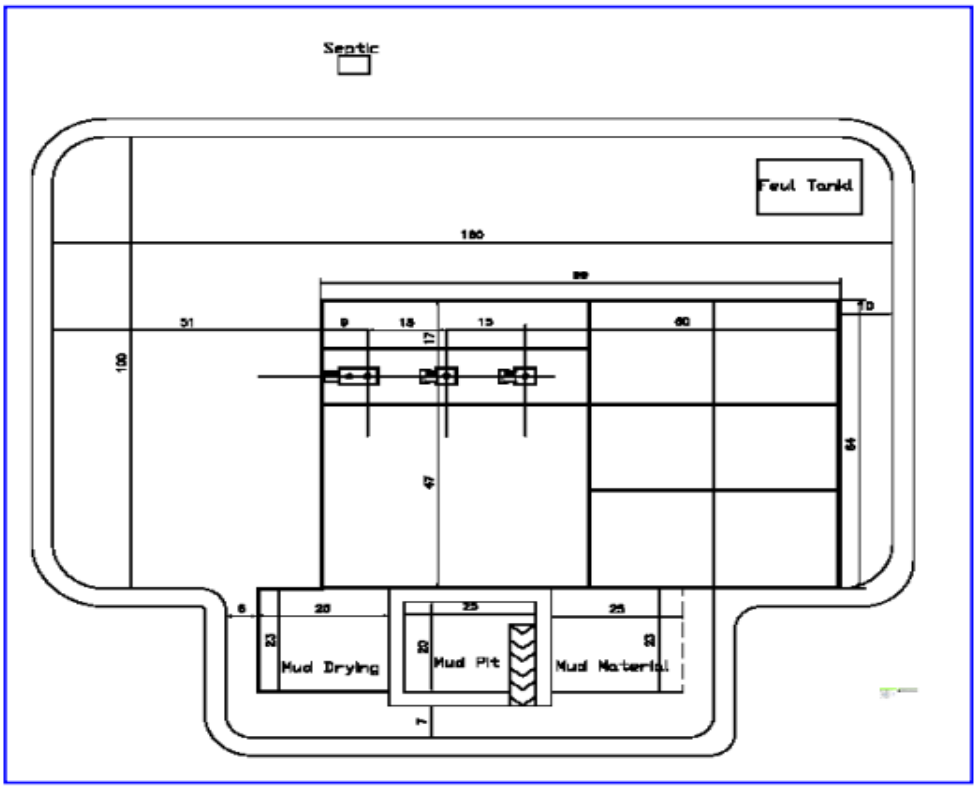
- One well in one pad.
- Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
- Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.



Well Pad for one well



Double-well Well Pad



Triple-well Well Pad

Figure 3: Well Pads Sketch

1.5 Reservoir Data

Reservoir Name:	Gadvan	Unit
Fluids:	Oil	….
Datum Depth:	3661	M,TVDss
Datum Pressure:	Gad-5c= 4915	Psi
Pressure Gradient:	0.3	Psi/ft
Temperature:	243.1	F
Temperature Gradient:	1.32 0.73 2.4	F/100ft C/100ft C/100m
H2S:	0	PPM
CO2:	25041	PPM
Salt Salinity :	200109	ppm
Calcium Content:		(mg/lit)

Reservoir	Sub-layer	Mid of Sub-Layer Depth in AZNS-075 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure
Gadvan	Khalij Memeber	-3616.5	241.1	1.32	6349	Water Pressure Gradient
	Gad-2	-3622.5	241.4		6359	Water Pressure Gradient
	Gad-3	-3628	241.6		6366	Water Pressure Gradient
	Gad-4	-3638	242.1		6380	Water Pressure Gradient
	Gad-5a	-3649	242.5		4854	Water Pressure Gradient
	Gad-5b	-3654	242.8		4859	Water Pressure Gradient
	Gad-5c	-3661	243.1		4915	Water Pressure Gradient
	Gad-6	-3677	243.8		4936	Water Pressure Gradient
	Gad-7	-3700	244.8		4971	Water Pressure Gradient
	Gad-8	-3749	246.9		5048	Water Pressure Gradient

1.6 Formation Pressure Prediction in Block III

Formation/Member	Interval	Pore Pressure	Collapse Pressure (Break-Out Pressure)	Fracture Pressure (Break-Down Pressure)
	m	SG	SG	SG
Aghajari	0~896	0.98~1.18	0.06~1.39	1.01~1.89
Gachsaran	896~1196	1.13~1.31	0.78~1.36	1.89~2.01
Asmari	1196~1380	1.14~1.26	0.6~1.28	1.43~1.79
Pabdeh/Jahrum	1380~2022	1.20~1.26	0.82~1.36	1.40~1.78
Gurpi	2022~2471	1.21~1.26	0.66~1.02	1.40~1.76
Ilam	2471~2546	1.18~1.24	0.2~0.9	1.41~1.72
Laffan	2546~2553	1.13~1.18	0.24~1.11	1.41~1.72
Sarvak	2553~3197	1.14~1.26	0.24~1.23	1.44~1.81
Kazhdumi	3197~3408	1.1~1.14	0.46~0.96	1.93~1.94
Dariyan	3408~3588	1.18~1.25	0.55~1.10	1.96~2.06
Gadvan	3588~TD	1.1~1.25	0.55~1.44	1.97~2.11

1.7 Well History

1.7.1 Drilling History

This well was drilled on 13 Jun 2015 by rig FATH-33 to produce oil from GADVAN formation as below operation summary:

1. 26” Hole was drilled, Ran & Cemented 20” Surface Casing. 20" CSG was run & cemented by inner stab-in stringer with slurry to surface.
2. N/U CHH and BOP Stack, Performed Pressure Test, Drilled out float shoe.
3. 17-1/2" hole was drilled. Run and cement 13-3/8” casing @ 1155.5m.
4. N/U CHS and BOP, Performed Pressure Test, drilled out float shoe.
5. 12 1/4" Hole was drilled. Run and cement 9 5/8" casing @ 2477.5m.
6. N/U THS and BOP Stack, performed pressure tests, drilled out float shoe.
7. 8 1/2" hole was drilled, 7" Liner was run & cemented from 2356m to 3799.5m. 7" L. Lap dry test was performed, no bubble was observed, 920 psi pressure build up was observed in the closing period.
8. Set two cement plugs & Rig was released on 30-Aug-2015.
9. Rig FATH-65 was spudded on 17 Jan 2017.
10. D/O Two cement plugs.
11. 7" Liner lap dry test was performed, no bubble was observed, 113 psi pressure build up was observed in closing period.
12. Perforation was performed inside of 7” liner, Perforation interval: 3659-3666 m & 3669-3678 m
13. Completion string was run and packer was set at 3575.34m.
14. Rig FATH-65 was released on 17-Feb-2017.
15. Acidizing was performed after releasing the rig. 30 bbls of 28% HCl was spotted by CT on perforation zone and then bullheaded into formation.
16. After unloading the well with diesel and nitrogen, it was observed the well was not flowing but annulus between production casing and tubing got pressure.
17. To check the status of reservoir interval tried to run Temperature, Pressure and Caliper logs in static and injection mode, but the operation of injection mode was not successful due to leakage in completion string and annulus pressure.

18. To test completion string and verify the leak point, plug & prong were set in SSD and string was tested by pressurizing, there was no leakage. The plug & prong were set in NO-GO landing nipple and tried to pressurize the string, pressure test was failed and it was confirmed that leak point is in interval between SSD and production packer.

19. It has been supposed water entry from Gadvan or even Kazhdumi formation into well is the main reason which has prevented the well flowing.

1.8 Well Montage before Workover#1

Field: South Azadegan
Pad no: 80
Location: Iran
Type: Vertical

Drilling F/Surface to TD		Perforation & Completion	
Rig No:	33 Fath	Rig No:	65 Fath
Spudded On:	13-Jun-15	Spudded On:	17-Jan-17
Rig Released:	30-Aug-15	Rig Released:	17-Feb-17
Rig Days:	77.75	Rig Days:	30.88

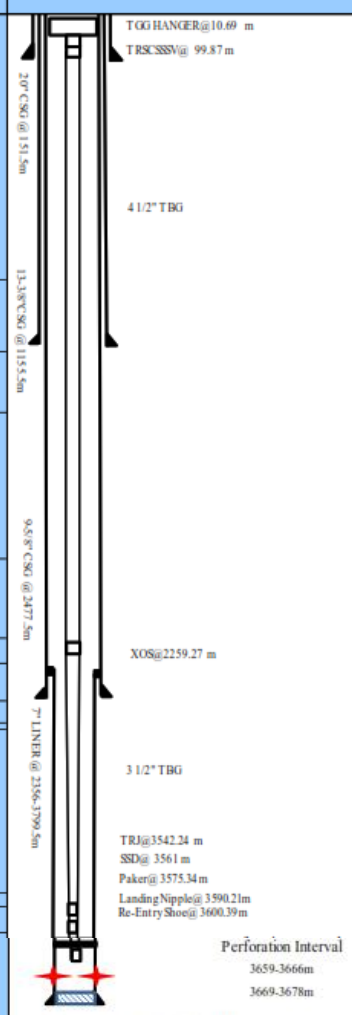
Moving after Suspention: 2.87 Days
Total Rig Days: 111.5

South Azadegan Oil Field AZNS-075 (G11)
as Drilled and Completed.

Coordinates: E: 186,427 N: 3,436,066
RTE: 12.95 / 13.55m
GLE: 3.95m
Final Well Status: Suspended In South Azadegan
PEDEC

Formation	Depth m,MD	Vertical Thick m
Aghajari	896 m	896 m
Gachsaran	1167 m	271 m
Asmari	1383 m	216m
Pabdeh	1939 m	556 m
U.Gurpi	2233 m	294 m
Tarbur	2333 m	100m
L.Gurpi	2471 m	138 m
Ilam	2546 m	75 m
Taffan	2552 m	6 m
Sar	3322 m	643 m
Kazh	3400 m	205 m
Dariyan		156 m

U-GAD: 3556 m MD
KhaiIUM: 3625 m MD
U-GAD-1: 3 /3525 MD
L-GAD: 3634 mMD





Bit Size/Type	ROP m/hr	Drilling Fluid	Remarks On Hole Condition	Casing	Well Head & BOP	Brief Cementing Report	Logs Tests	Drilling History
26" TSK/115/SS-TMZGC (1)	6.50	BENTONITE MUD, MW:68PCF	NL-NG	20" K-55, 94ppf, BTC	Wellhead C.H.H(20 3/4"*20"3K)	CMT Class "G". S.W: 115pcf, S.V: 275 bbl) Return CMT @ surface.	N/A	26" Hole Section: This section was drilled from 12.5 to 152 m with 68 pcf, Bentonite mud. 20" casing was run (K-55, 94ppf, BTC) to 151.5 m & cemented. cement slurry was returned to surface. C.H.H (20-3/4" 3K) was installed.
17 1/2" TSK/135/MS-MZG (2)	8.50	BENTONITE MUD, MW:68PCF	Tight Hole, Increase loss to 40 BPH, Increase CA++, Bit Balling	13-3/8" N-80 68ppf BTC	Wellhead C.H.S (20 3/4" 3K * 13-5/8" 5K) BOP 20-3/4" 3K	Lead Cmt: Cement class "G" SW: 93pcf SV:814bbls Tail Cmt: Cement Class "G" SW: 115pcf SV: 134bbls 300 bbls CMT slurry Return @ Surface	CDR/GR	17 1/2" Hole Section: This section was started from 152 to 300m. with Bent. Mud (68pcf). In 300m, Ben. mud was replaced by 69 pcf KCL-Polymer mud. MW was increased F/69 to 74 gradually. Drilling was continued to 1156m. During drilling (502m), Due to 15-40 BPH Losses, Pumped LCM pill and decrease loss to 15 BPH, During drilling (1070-1156m), Due to increase CA++, MW was increased to 79 pcf. While hole condition for logging W&R some intervals such as (695-705, 850-875, 990-1015), CDR/GR was performed F/966 to 100m & Try to pass logging tools F/966 m, no success. 13-3/8 CSG was run to 1155.5m. Pumping cement & 300 bbl cement slurry was returned to surface. Bumping was not observed. Check back flow, OK. 20 3/4" 3K, 13 5/8" 5K C.H.S was installed. Run 12 1/4" bit and tag TOC @ 1053.5m. D/O cement plug, float collar to 1148m. Perform casing integrity test to 1500 psi,ok. Drill out to 1153 m. Decrease and balance mud weight to 75pcf. Drill out to 1156 m.
17 1/2" KINGDREAM/M323/KM166 3SGRA(3)	5.40	KCL/POLY. MUD, MW:69-79PCF						
12 1/4" TSK/437/M10GFP(4)	1.5	KCL/POLY. MUD, MW:75-76PCF	Heavy loss, Tight hole	9-5/8" L-80 47ppf PJE	Wellhead T.H.S (13-5/8" 5K* 11" 10K BOP 13 5/8" 10K	Lead Cmt: Cement class "G" SW: 85pcf SV:809bbls Tail Cmt: Cement Class "G" SW: 115pcf SV: 75bbls 50 bbls contaminated & 120 bbls net CMT slurry Return @ Surface	CDR/SONIC/GR - CBL/VDL/CCL/GR	12 1/4" Hole Section: 12-1/4" hole section was started from 1156m with 75 pcf KCL Polymer mud. FIT was performed @ 1159m & EMW was 85 pcf. During drilling (1329-1533m), Increase mud weight F/75 to 76 pcf Due to tight hole. While drilling @ 1530m detect 130 BPH losses, Try to control loss by LCM while drilling to 1800m, no success. Perform Sodium Silicate & Cement job and control loss successfully. Drilling was continued to 2478m (Section TD). After condition trip,GR/SONIC/CDR log was perormed F/2478 TO 1155m. 9-5/8" casing string was run from surface to 2477.5m. Mixing & pumping CMT, Displace cement and Bumping was observed (1500 psi). Inc. pressure to 3000 psi for 15 min as CIT. Check back flow, OK. 50 bbls contaminated & 120 bbls net CMT slurry Return @ Surface. 13-5/8" 5K X 11" 10K T.HS was installed. Run 8 1/2" bit and tag TOC @ 2452m. Drill out cement float collar & cement below the float collar to 2471m. Displace 76 pcf KCL Polymer mud W/76 pcf Glycol KCL Polymer mud. Drill out to 2478m. CBL/VDL/CCL/GR Log was performed from 2477.5 to 1100m.
12 1/4" KINGDREAM/M223/KM166 3SGRA(5)	10.7							
12 1/4" VAREL/537/HE34DJMRSV(6)	7							
12 1/4" KINGDREAM/M223/KM166 3SGRA(5*R2)	5.30							
8 1/2" VAREL/217/DT4DGMRS (7)	1.20	KCL/POLY/ GLY. MUD, MW:76-82 PCF	Tight Hole, Flow the well, Stuck Logging tools	7" Liner L-80 29ppf HSM1	BOP 13 5/8" 10K	Lead Cmt: Cement class "G" SW: 88pcf SV:210bbls Tail Cmt: Cement Class "G" SW: 115pcf SV: 29bbls	TTMR/TELC/NGRS/MSFL/DLL/DAS - CNL/DSL In case/ hole - CBL/VDL/CCL/GR	8 1/2" Hole Section: 8 1/2" hole section was started from 2478 with 76 pcf Glycol/KCL/Polymer mud. FIT was performed @ 2487.5m & EMW was 88 pcf.While drilling F/3666 to 3671m, Increase mud weight to 77 pcf due to thight hole. While drilling in 3743m, Flow the well (60 BPH Gain), Kill the well by 82 pcf mud. Drilling was continued to 3800m (Section TD). After hole condition run logging as TTMR/TELC/NGRS/MSFL/DLL/DAS to 3260m, Took weight, Try to pass this point, no success. Logging tools stuck while POOH to 3200m and got free with 5000 Lbs O.Pull. Repeat hole condition and perform previouse log from 3800 to 2477.5m. Ran logging as LDT/CNT/NGRS/TELC/TTMR to 3740m, took weight, Try to pass this point, no succ & LDT was failed. 7" liner was run from 2356m to 3799.5m (Liner shoe). After mixing & pumping CMT Shearing observed with 2200psi, but Bumping was n't observed. Run 8 1/2" bit and took weight @2330m. Try to wash down, no success. Drill out cement F/2330 to 2356m(7" L/Lap). POOH & P/U 6-1/8" bit to 2358m, D/O P.O.Bushing. RIH to 3736m, Took weight. D/O Plugs and hard cement to 3765m (7" LDC). Positive test 7" L.Lap to 2000 psi, ok. Perform 7" L.Lap dry test with no bubble and 920 psi pressure build up in closing period. Perform CNL/DSL & CBL/VDL/CCL/GR Log. Due to shortage of completion string & Acc. suspend the well with two cement plugs (Plug#1:2144m-2256m, Plug#2:1394-1500m). Rig was released @ 30-Aug.-2015 on 10:00.
8 1/2" REED/M432/DSZX-613M (8)	6.80							
		KCL/POLY/ GLY. MUD, MW:82 PCF	No Problem		Wellhead X-Mass Tree (11" 5K*4 1/16 10K) BOP 13 5/8" 10K	No Cement	Y/N	Re-Entry Operation: This Operation was performed for D/O CMT plugs, Perforation & Completion. After N/U BOP 10K, 13 5/8, D/O CMT plugs number 2 and one (mentioned in the last hole) was performed with 8 1/2" bit(100,88 pcf). RIH to 7" liner lap. POOH to surface. Change BHA and RIH for scrape inside 7" liner to 3247 m and took weight. RIH with pumping and rotation to 3256m. POOH & L/D scraper.TD check with 6-1/8" bit while W & R from 3256 to 3766 m (LDC Depth). Displace the well with 75 pcf S.S.Water. POOH to 7" L.Lap and Perform 7" Liner integrity test to 3600 psi. 7" L.Lap dry test with no bubble in opening situation and 113 psi pressure build up in closing situation. Perforate 3659-3666 m & 3669-3678m as TCP program. Check perforated interval with CCL/GR log,ok. Complete the well by 4-1/2"*3-1/2" , 10K completion string as completion program. Completion string was run & every 50 Jts pressure test (6000 psi) was performed (Any plug in NO-GO). Ret check valve from SSD. Close SSD.Circulate fresh complertion fluid as 1.5 string cap. with pressure less than 150 psi. Set WRB-2 check valve in L.Nipple after 2 times. Space out completion string and set PKR. Test PKR & TBG hanger to 2000 psi , ok. Completion string test to 6000 psi. Retrive check valve from L.Nipple after 2 times. Pressure test control line to 8000 psi. Close 3SV. Install TWCV in TBG Hanger. Pressure Test TWCV (6000 psi), ok. N/D BOP Stack 13 5/8", 10K. install Bottom master valve, test it, OK. Test C/L (8000 psi). Install 4-1/16",10K X-Mass tree and pressure test to 7500 psi,ok. TWCV in tubing hanger. Released Rig @ 06:00 on 17-Feb-2017.

1.9 Casing & Liner Depth & Data

Casing Size (in)		Interval(m)	Weight (ppf)	Grade	Connection	Optimum Torque (ft.lbs)	Collapse (Psi)	Burst (Psi)
OD	ID							
20	19.124	0-151.5	94	K-55	BTC	Triangle	520	2110
13.375	12.415	0-1155.5	68	N-80	BTC	Triangle	2260	5020
9.625	8.68	0-2477.5	47	L-80 (1)	PJE	11500	13740	15110
7	6.184	2165-3728	29	L-80 (1)	HSM-1	10200	8530	11220
5	4.276	3628-3888.5	18	L-80 (1)	HSG3-FJ	6505	10487	10131

1.10 Well Schematic before Workover#1

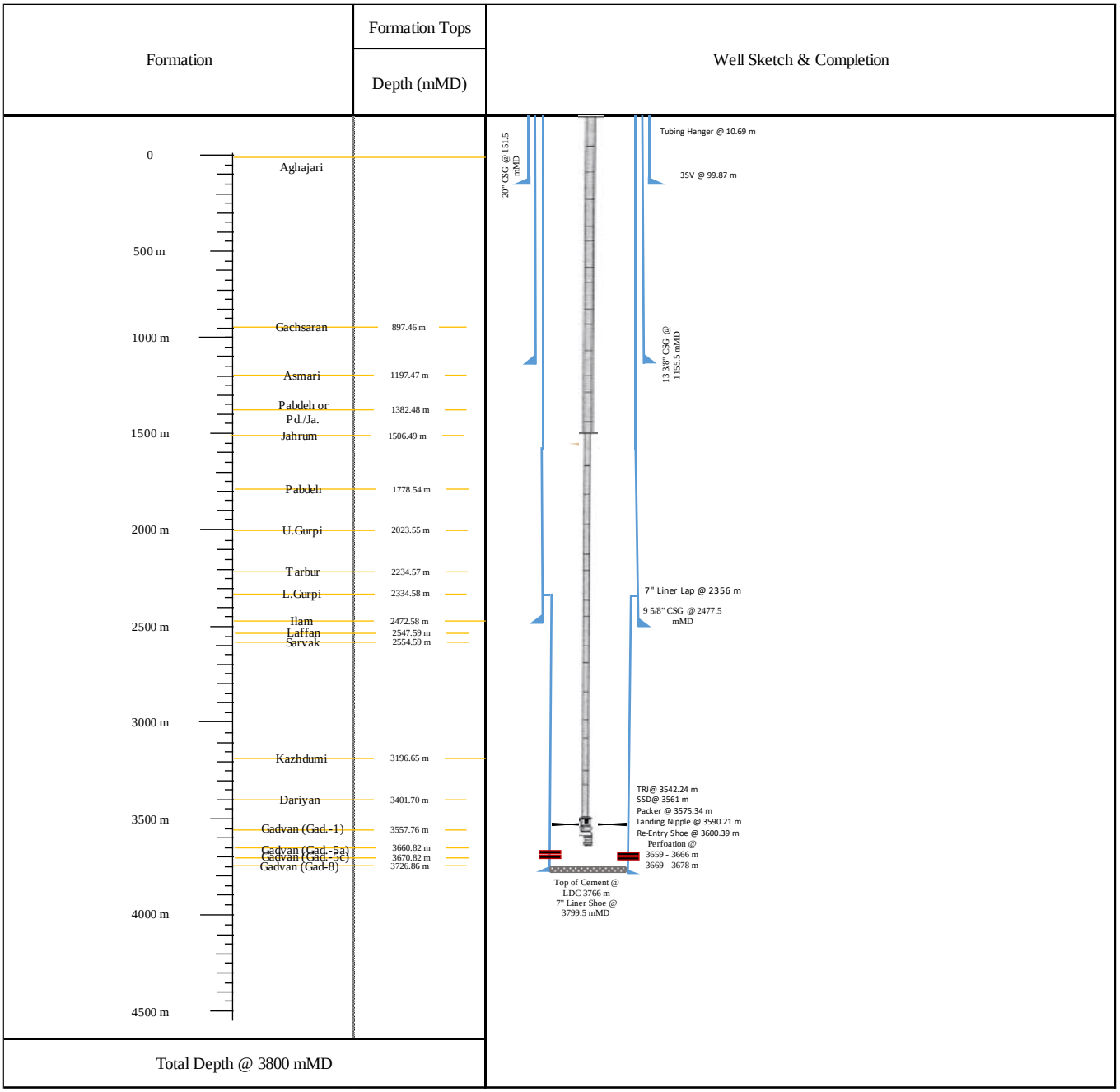


Figure 4: Well Schematic Before workover #1

1.11 Well Schematic after Workover#1

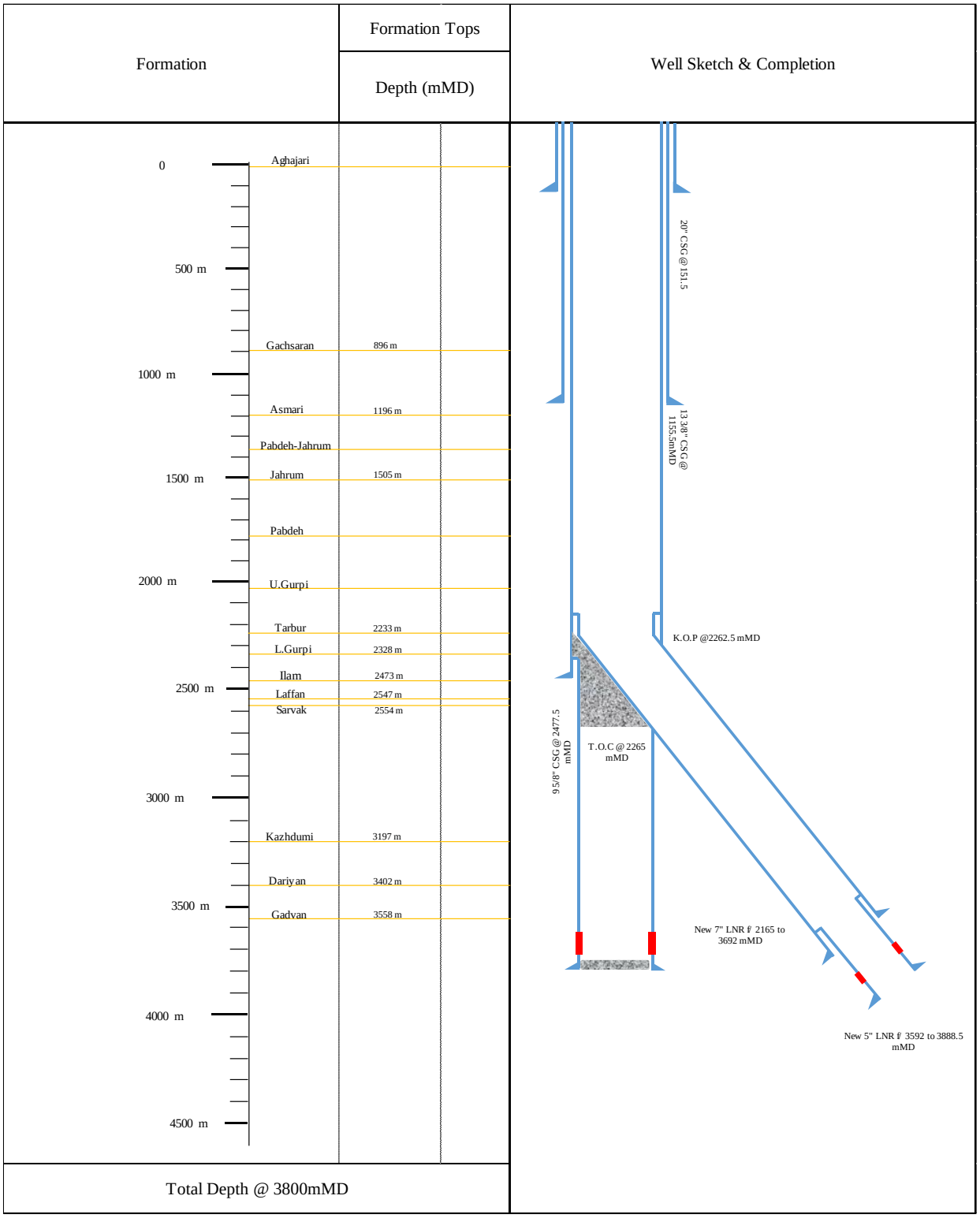


Figure 5: Well Schematic After workover #1

Geological Information

2.1 Formation Stratigraphy and Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
Miocene	Fars	Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi Tarbur L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
	Bangestan	Laffan		Claystone	2691	2642	2625
Turonian Cenomanian	Bangestan	Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian	Bangestan	Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian	Khami	Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian	Khami	Fahliyan		Limestone, Marl	4092	3989	3974

2. Logging Program

Hole Size	Casing / Liner	Open Hole	Cased Hole Wire Line Logging
8 3/8	7” Liner	MWD, Full Suite log, GR	SBT, VDL/CCL/GR
6 1/8"	5” Liner	MWD, Full Suite log, CDR-GR *	RBT or CAST/CBL/VDL Perforation CCL-GR

3. Program Outline

4.1 Contractors Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 201
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	SEALAND
5	Mud Logging	Pars Geo Data
6	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC
15	Directional Drilling	HANA ENERGY

4.2 Time Breakdown

AZNS-075 Work over Time Break Down	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over												
Held PJSM. Spud the Rig. Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC.	Spud		SV	3800		0	0.00	0.0	0.0	0.0	0.00	0.0
Check and record all wellhead pressures.	Wellhead		SV	3800		18	0.75	0.8	0.8	19.8	0.83	0.8
R/U Well Test lines from XMT to burning pit. and connect f/ 4 ½"-9 5/8" annulus side out let valve to Pump Truck/mud pumps.	Well Test		SV	3800		18	0.75	1.5	0.8	19.8	0.83	1.7
R/UP Slick Line & Test Lubricator,Open sub surface safety valve, install Protection Sleeve. Note:Suitable Protection Sleeve & Running-Pulling Tools Shall be Provided by Client. If wellhead pressure is not Zero, Contractor shall inform Client. Note: if Retriving 3SV is not Possible Remedial plan to be prepared in advance by Operators and will be followed after client's confirmation	Trip		SV	3800		36	1.50	3.0	1.7	39.6	1.65	3.3
Release well pressure through Burning Line in Safe Mode to Burning Pit & Also Monitor W.H.Pressure.	Well Test		SV	3800		36	1.50	4.5	1.7	39.6	1.65	5.0
RIH w/SL & Run Shifting Tools Try to Open SSD @ 3562 Mt, POOH to Surf., R/D Equip. Note:if its not possible to open SSD follow Contingency Plan#01	Trip		SV	3800		36	1.50	6.0	1.7	39.6	1.65	6.6
Displace the well with proper mud from 9 5/8" x 4 1/2" Ann. & Circ. To get uniform mud. Monitor the we ll. Note: if the completion string was wash out from other depths follow Contingency Plan#02	Trip		SV	3800		60	2.50	8.5	2.8	66.0	2.75	9.4
R/D well test lines. Set BPV and Nipple down Xmas tree, Nipple up BOP and test same. Note:if Unable to set BPV Due to Damaged Profile, Install 3SV Blanking plug or 7 inch Wire Line Bridge Plug will be followed after client's confirmation	Wellhead		SV	3800		72	3.00	11.5	3.3	79.2	3.30	12.7
RIH with TBG hanger retrieving tool and engage with TBG hanger. Note:if Tubing Hanger Thread damaged, RIH w/Spear & Try to engage w/Tubing Hanger will be followed after client's confirmation	Wellhead		SV	3800		72	3.00	14.5	3.3	79.2	3.30	16.0
Unlatch Seal Nipple from permanent Packer and pull completion string out of hole (from 3575m). Note:if unable to unlatch completion string due to string pack off with huge amount of debries follow contingency plan#03 Note:if unable to unlatch completion string due to string wash out follow contingency plan#04 Note:if observe any damage on THS follow contingency plan#05	Trip		SV	3800		144	6.00	20.5	6.6	158.4	6.60	22.6
RIH with 6-1/8" Mill to 3575m (Top of permanent Packer),Clear The Well CBU, POOH.	Trip		SV	3800		72	3.00	23.5	3.3	79.2	3.30	25.9
M/U Tandem scraper BHA to scrap 9 5/8" casing and inside 7" liner (To top of Permanent Packer), CBU, POOH 7" scraper BHA and L/D same.	Trip		SV	3800		0	0.00	23.5	0.0	0.0	0.00	25.9
RIH with Wire line Corrosion Logging Tools to Top of Permanent Packer bottom, log up to surface	Trip		SV	3800		48	2.00	25.5	2.2	52.8	2.20	28.1
RIH with OEDP to 2340 m , Perform cement plug job from 2340 to 2240 m,WOC to get hard,RIH & Tag CMT, POOH to surface.	CMT		Da.	2240		60	2.50	28.0	2.8	66.0	2.75	30.8
RIH w/ 6 1/8" clean out BHA to TOC, D/O cement to 2265 m (a minimum of 5m hard cement need to be drilled out), CBU, POOH to surface and L/D 6 1/8" Drill out BHA. Note: For test Cement plug follow contingency plan#6 Note: For dry test 7" Liner lap follow contingency plan#7	D/O		Da.	2265		60	2.50	30.5	2.8	66.0	2.75	33.6

RIH w/ 9 5/8" Scraper and scrape inside 9 5/8" Casing (Whipstock setting depth), CBU, POOH to Surf.	Trip		Da.	2265		0	0.00	30.5	0.0	0.0	0.00	33.6
RIH RTTS PKR Inside 7" liner & Perform plugs dry test, POOH to surface. Note: If dry test is not ok, follow contingency plan #6.	Dry Test		Da.	2265		0	0.00	30.5	0.0	0.0	0.00	33.6
RIH RTTS PKR & perform 7" liner lap & plugs dry test, POOH to surface. Note: If dry test is not ok, follow contingency plan #7.	Dry Test		Da.	2265		0	0.00	30.5	0.0	0.0	0.00	33.6
Run and set 9 5/8" Whip stock at 2265m, open window, drill 3m new formation. CBU. POOH.	Side Track	3	Da.	2268		96	4.00	34.5	4.4	105.6	4.40	38.0
P/U & M/U New 8-3/8" Tri-Cone Bit and Directional BHA (LWD,MWD), RIH to Btm.	Trip			2268		18	0.75	35.3	0.8	19.9	0.83	38.8
Continue Directional Drilling Formation to 2318.0 Mt.	Drilling	50		2318	2.5	20	0.83	36.1	0.9	22.0	0.92	39.7
Perform Hole condition POOH to surface.	Hole Cond/Circ			2318		20.0	0.83	36.9	0.9	22.0	0.92	40.6
Test BOP Stack and all Related Lines as NISOC POLICY	BOP Test			2318		6	0.25	37.2	0.3	6.6	0.28	40.9
P/U & M/U New 8-3/8" PDC Bit and Directional BHA, RIH to Btm.	Trip			2318		18	0.76	37.9	0.8	20.1	0.84	41.7
Continue Directional Drilling Formation to 2518.0 Mt.	Drilling	200		2518	2.5	80	3.33	41.3	3.7	88.0	3.67	45.4
CBU, Perform Wiper trip, RIH back to btm.	Hole Cond/Circ			2518		8	0.33	41.6	0.4	8.8	0.37	45.8
Continue Directional Drilling Formation to 2718.0 Mt.	Drilling	200		2718	2.5	80	3.33	44.9	3.7	88.0	3.67	49.4
Perform Hole condition POOH to surface.	Hole Cond/Circ			2718		24	0.99	45.9	1.1	26.3	1.09	50.5
P/U & M/U 8-3/8" PDC Bit and Directional BHA, RIH to Btm.	Trip			2718		24	1.00	46.9	1.1	26.4	1.10	51.6
Continue Directional Drilling Formation to 2918.0 Mt.	Drilling	200		2918	2.5	80	3.33	50.3	3.7	88.0	3.67	55.3
CBU, Perform Wiper trip, RIH back to btm.	Hole Cond/Circ			2918		8	0.33	50.6	0.4	8.8	0.37	55.7
Continue Directional Drilling Formation to 3118.0 Mt.	Drilling	200		3118	2.5	80	3.33	53.9	3.7	88.0	3.67	59.3
Perform Hole condition POOH to surface.	Hole Cond/Circ			3118		26	1.08	55.0	1.2	28.6	1.19	60.5
Test BOP Stack and all Related Lines as NISOC POLICY	BOP Test			3118		6	0.25	55.3	0.3	6.6	0.28	60.8
P/U & M/U New 8-3/8" PDC Bit and Directional BHA, RIH to Btm.	Trip			3118		24	1.00	56.3	1.1	26.4	1.10	61.9
Continue Directional Drilling Formation to 3318.0 Mt.	Drilling	200		3318	2.5	80	3.33	59.6	3.7	88.0	3.67	65.6
CBU, Perform Wiper trip, RIH back to btm.	Hole Cond/Circ			3318		8	0.33	59.9	0.4	8.8	0.37	65.9
Continue Directional Drilling Formation to 3518.0 Mt.	Drilling	200		3518	2.5	80	3.33	63.3	3.7	88.0	3.67	69.6

Perform Hole condition POOH to surface.	Hole Cond/Circ			3518		30	1.25	64.5	1.4	33.0	1.38	71.0
P/U & M/U 8-3/8" PDC Bit and Directional BHA, RIH to Btm.	Trip			3518		24	1.00	65.5	1.1	26.4	1.10	72.1
Continue Directional Drilling Formation to Section TD 3728.0 Mt.	Drilling	210		3728	2.5	84	3.50	69.0	3.9	92.4	3.85	75.9
Perform Hole condition perior to run log, Spot pill, POOH to surface.	Hole Cond/Circ			3728		32	1.33	70.3	1.5	35.2	1.47	77.4
Test BOP Stack and all Related Lines as NISOC POLICY	BOP Test			3728		6	0.25	70.6	0.3	6.6	0.28	77.7
Install 7" Shooting Nipple and test same, R/U WL/TLC and perform reservoir Full Suit Logs, then POOH.	Log			3728		42	1.75	72.3	1.9	46.2	1.93	79.6
P/U 8-3/8" Bit and BHA, RIH and Perform condition trip for 7" liner running, CBU till shaker got clean, Pump and Spot Hi-Vis Pill, POOH.	Hole Cond/Circ			3728		48	2.00	74.3	2.2	52.8	2.20	81.8
R/U CSG running Equipment, Run 7" Liner to Btm	CSG			3728		48	1.99	76.3	2.2	52.6	2.19	84.0
Perform 7" Liner cementing operation.	CMT			3728		8	0.33	76.7	0.4	8.8	0.37	84.3
POOH w/ L.Hanger setting tool to 2165m, Flush string & CBU.	Trip			3728		4	0.17	76.8	0.2	4.4	0.18	84.5
POOH to surface & L/D L.Hanger setting tool.	Trip			3728		8	0.33	77.2	0.4	8.8	0.37	84.9
P/U & M/U 8-3/8" MT Bit BHA, RIH to L.Lap @ +/- 2165 m while D/Out cement (If any cement: Considered 7STD wet for CMT Job), Circ & clean the well, POOH.	Trip			3728		32	1.33	78.5	1.5	35.2	1.47	86.4
P/U & M/U 6-1/8" MT Bit BHA, RIH to top of cement, D/O POB and clean out inside liner to top of LDC. Perform CIT to 3000 psi, CBU, POOH to surface.	D/O			3728		48	2.02	80.5	2.2	53.2	2.22	88.6
P/U & M/U 9 5/8" Scrapper with same BHA and 8 3/8" R.Bit, RIH To 20 m above Liner Lap. Scrap the Dry test packer setting Depth	Dry Test			3728		30	1.25	81.8	1.4	33.0	1.38	89.9
P/U dry test equipment and run to 20 m above L.Lap, Perform 7" L.Lap dry test as per program. Secure the well, POOH to Surface and L/D assemblies.	Dry Test			3728		96	4.00	85.8	4.4	105.6	4.40	94.3
RIH W/ 6 1/8" Bit While P/U 3 1/2" and 5" DPs as tandem BHA. Continue Clean Out Inside Liner to 5 m above shoe.check and report CMT hardness before performing SBT Log. CBU, POOH to surface.	D/O			3728		56	2.33	88.1	2.6	61.6	2.57	96.9
P/U 7" casing Scraper & Perform Scraper Trip inside 7" liner Prior to Run SBT Log.	Log			3728		30	1.25	89.4	1.4	33.0	1.38	98.3
R/U Logging Equipment.Install Shooting Nipple, Run Segmented Bond Tools inside 7" Liner.	Log			3728		24	1.00	90.4	1.1	26.4	1.10	99.4
RIH W/ 6 1/8" R.Bit (Mill) While P/U 3 1/2" and 5" DPs as tandem BHA. Drill out 7" liner shoe to total Depth (TD), Commence Drilling 5 m of new section, CBU,Perform FIT, displace the hole with new mud, POOH to surface.	D/O			3728		50	2.08	92.4	2.3	55.0	2.29	101.7

P/U & M/U New 6-1/8" Rock Bit and Directional BHA(LWD,MWD), RIH to Btm.	Trip		Da.	3728		30	1.25	93.7	1.4	33.0	1.38	103.1
Continue Directional Drilling Formation to Section TD 3888.5 Mt.	Drilling	160.5	Da/Ga.	3888.5	3	54	2.23	95.9	2.5	58.9	2.45	105.5
Perform Hole condition POOH to surface. Note:for Reservoir Full Suite Log & Scraper trip for 5" Liner hanger seting area follow contingency plan#8	Hole Cond/Circ		Ga.	3888.5		60	2.50	98.4	2.8	66.0	2.75	108.3
R/U WL/TLC and perform reservoir Full Suit Logs, then POOH.	Log		Ga.	3888.5		0	0.00	98.4	0.0	0.0	0.00	108.3
RIH w/ 7" Scraper clean out BHA and scrape around 3628 m for Liner Hanger setting area, POOH.	Hole Cond/Circ		Ga.	3888.5		0	0.00	98.4	0.0	0.0	0.00	108.3
Perform condition trip for 5" liner running (5" flush connection), CBU, POOH.	Hole Cond/Circ		Ga.	3888.5		0	0.00	98.4	0.0	0.0	0.00	108.3
R/U CSG running gear, Run 5" Liner to Btm, and hang same with 7" x 5" liner hanger packer at 3628m (100 m overlap)	CSG		Ga.	3888.5		60	2.50	100.9	2.8	66.0	2.75	111.0
Perform 5" Liner cementing operation.	CMT		Ga.	3888.5		20	0.83	101.7	0.9	22.0	0.92	111.9
POOH w/ L.Hanger setting tool to 3000m, Flush string & CBU.	Trip		Ga.	3888.5		18	0.75	102.5	0.8	19.8	0.83	112.7
POOH to surface & L/D L.Hanger setting tool.	Trip		Ga.	3888.5		24	1.00	103.5	1.1	26.4	1.10	113.8
P/U & M/U 6-1/8" MT Bit BHA, RIH to L.Lap @ +/- 3628m while D/Out cement (If any cement: Considered 7STD wet for CMT Job), Circ & clean the well, POOH.	Trip		Ga.	3888.5		72	3.00	106.5	3.3	79.2	3.30	117.1
L/D excess 5" DPs/HWDP.	Trip		Ga.	3888.5		0	0.00	106.5	0.0	0.0	0.00	117.1
P/U & M/U 4-3/4" DCs & 3-1/2" DPs/HWDPS, R/Back same in derrick.	Trip		Ga.	3888.5		0	0.00	106.5	0.0	0.0	0.00	117.1
P/U & M/U 4-1/8" MT Bit BHA, RIH to tag POB, D/Out POB, RIH to LDC depth @ +/- 3855m while D/Out any cement, work on junk sub and CBU. Perform 5" Liner CIT. (80% of Burst press) to 2000 psi,for 15 min.CBU, POOH to Surface. Note:for dry test 5" Liner lap & Perform CMT quality Logging of 5" Liner follow contingency plan#9	D/O		Ga.	3888.5		72	3.00	109.5	3.3	79.2	3.30	120.4
RIH with 4-1/8" MT Bit and 7" scraper to 3620 m, CBU, POOH to Surface.	Trip		Ga.	3888.5		0	0.00	109.5	0.0	0.0	0.00	120.4
RIH with 7" RTTS & Dry test BHA, RIH to L.Lap @ +/- 3628m, Perform 5" Liner lap Dry Test (based on Gadvan Reservoir Datum Pressure), POOH.Note: If dry test is not ok, follow contingency plan #09.	Dry Test		Ga.	3888.5		0	0.00	109.5	0.0	0.0	0.00	120.4
Change Ram, Install shooting nipple, R/U WL/TLC Logging Eq. Perform CMT quality Logging of 5" Liner from 5" liner LDC @ 3855 m to 7" Liner Lap @ 2165 m (RBT/VDL/CCL/GR),POOH to Surface.	Log		Ga.	3888.5		0	0.00	109.5	0.0	0.0	0.00	120.4
P/U & M/U TCP Assy, RIH to perforation depth, correlate depth with WL and perforate Guns as program.POOH to Surface.	Perforation		Ga.	3888.5		120	5.00	114.5	5.5	132.0	5.50	125.9
RIH with GR-CCL TLC log to confirm perforation intervals, POOH.	Log		Ga.	3888.5		0	0.00	114.5	0.0	0.0	0.00	125.9
RIH scraper and scrape 7" production packer setting area, displace the well with Calcium Chloride (CaCl2) as completion fluid, POOH to surface.	Trip		Ga.	3888.5		60	2.50	117.0	2.8	66.0	2.75	128.7
RIH w/ DP's and DC's, POOH & L/D same.	Trip		Ga.	3888.5		0	0.00	117.0	0.0	0.0	0.00	128.7

Run 7.5K completion string as per program.	Completion		Ga.	3888.5		144	6.00	123.0	6.6	158.4	6.60	135.3
Install Plug and Prong inside No-Go ,Set BPV inside TBH, N/D BOP's & install new XMT.	Completion		Ga.	3888.5		60	2.50	125.5	2.8	66.0	2.75	138.0
Retrive Plug and Prong and BPV, install TWCV and Pressure test the XMT.	Completion		Ga.	3888.5		18	0.75	126.2	0.8	19.8	0.83	138.9
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.	Completion		Ga.	3888.5		24	1.00	127.2	1.1	26.4	1.10	140.0
Retrieve TWCV,R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
RIH with S/L, retrieve protection sleeve.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion		Ga.	3888.5		Rig Less	0.00	0.0	0.0	0.0	0.00	140.0
Total Duration of 6-1/8" Hole Section:							***127.2***	***127.2***			***140.0***	***140.0***
NOTE:												
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip			3888.5		12	0.50	0.5	0.6	13.2	0.55	140.5
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service			3888.5		70	2.92	3.4	3.2	77.0	3.21	143.7
Total Duration of AZNS-075 Work-over Well:								3.4				143.7

Contingency Plans TBD of Well: AZNS-075 Work-over	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Based on operation situation in each step, the following contingency plans might be followed												
Contingency Plan#01 R/U & RIH W/Coliding Tools & Colide Completion String from Proper Depth,POOH & L/D	Completion		SV	3888.5		288	12.00	12.00	13.2	316.8	13.20	13.2
Contingency Plan#02 Displace well with heavy mud and try to kill the well,Circulate & Balance MW.	Completion		SV	3888.5		192	8.00	20.00	8.8	211.2	8.80	22.0
Contingency Plan#03 R/U & RIH with FPIT tools to check string free point,POOH to Surface & L/D.R/U & RIH W/Coliding Tools & Colide Completion String from Proper Depth,POOH & L/D.POOH Completion String from Coliding Depths in Several Runs.	Completion		SV	3888.5		288	12.00	32.00	13.2	316.8	13.20	35.2
Contingency Plan#04 Pull Completion String untill string will be part from weaked point then POOH & L/D String in Several attempt.Mill the remainder of fish & packer.	Completion		SV	3888.5		750	31.25	63.25	34.4	825.0	34.38	69.6
contingency plan#05 if observe any leakage and or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG). Note:Proper Equipment shall be provided by client and install by contractor.	Wellhead		SV	3888.5		320	13.33	76.58	14.7	352.0	14.67	84.2
contingency plan#06 Perform Scraper trip. Perform cement plugs test, POOH to surface. Note: If test is not ok, RIH and set another plug at 3420m, POOH will be followed.	Bridge Plug		SV	3888.5		300	12.50	89.08	13.8	330.0	13.75	98.0
contingency plan#07 RIH RTTS PKR & perform 7" liner lap dry test, POOH to surface. Note: If dry test is not ok, P/U & M/U 7" SN Packer, RIH and set a 7" SN Packer, POOH to Surface will be followed.Stand Back required DPs, HWDPs and DC.	Trip		SV	3888.5		296	12.33	101.42	13.6	325.6	13.57	111.6
contingency plan#08 Perform condition trip R/U WL/TLC and perform reservoir Full Suit Logs, then POOH. Perform Scraper Trip for 5" Liner Hanger setting area & POOH.Stand Back required DPs, HWDPs and DC. Stand Back required DPs, HWDPs and DC.	Trip		SV	3888.5		402	16.75	118.17	18.4	442.2	18.43	130.0
contingency plan#09 Perform Scraper Trip, RIH with 7" RTTS & Dry test BHA, RIH to L.Lap @ +/- 3628m, Perform 5" Liner lap Dry Test (based on Gadvan Reservoir Datum Pressure), POOH. If dry test is not ok,P/U & M/U 5" SN Packer, RIH and set a 5" SN Packer on Top of Liner, POOH to Surface will be followed .Stand Back required DPs, HWDPs and DC. Change Ram, Install shooting nipple, R/U WL/TLC Logging Eq. Perform CMT quality Logging of 5" Liner from 5" liner LDC @ 3855 m to 7" Liner Lap @ 2165 m (SBT/CBL/VDL),POOH to Surface. RIH with GR-CCL TLC log to confirm perforation intervals, POOH.	Trip		Ga.	3888.5		419	17.46	135.63	19.2	460.9	19.20	149.2
Total Duration of Contingency Plan AZNS-075 Work-over Well:								135.6				149.2

4. Sequence of Operation

5.1 Main Objective

Based on performed studies, the completion string should be pulled out, the well should be sidetracked then the well should be re-completed

5.2 Main plan Operation in Presence of Drilling Rig

Based on performed studies, the completion string should be pulled out, the well should be side tracked and then the well should be re-completed. The sequence of workover operation will be as following:

1. Held PJSM. Spud the Rig as per PSSR. Stand back required 78 Std 5" Dps, 50 Std 3-1/2" DPs and 6-1/2" DC.
2. Check and record all wellhead pressures.
3. R/U Well Test lines from XMT to burning pit. and connect f/ 4 ½”
   -9 5/8” annulus side out let valve
   To Pump Truck/mud pumps.
4. R/UP Slick Line & Test Lubricator, retrieve sub surface safety valve, install dummy safety valve.
   Note: Suitable Dummy Safety Valve Shall be provided by Client. If wellhead pressure is not Zero, Contractor shall inform Client.
   Note: if retrieving 3SV is not Possible Remedial plan to be prepared in advance by Operators and will be followed after client’s confirmation
5. Release well pressure through Burning Line in Safe Mode to Burning Pit & Also Monitor W.H. Pressure.
6. RIH w/SL & Run Shifting Tools Try to Open SSD @ 3562 Mt, POOH to Surface, R/D Equip.
7. R/U SL, RIH & Try to open SSD, POOH to Surface.
   Note: if it's not possible to open SSD follow Contingency Plan#01
8. Displace the well with proper mud from 9 5/8" x 4 1/2" Ann. & Circ. To get uniform mud. Monito r
   The well. Install BPV in Tubing Hanger.
   Note: if the completion string was wash out from other depths follow Contingency Plan#02
9. RIH with TBG hanger retrieving tool and engage with TBG hanger.
   Note: if Tubing Hanger Thread damaged, RIH w/Spear & Try to engage w/Tubing Hanger will be followed after client’s confirmation
   Unlatch Seal Nipple from permanent Packer and pull completion string out of hole (from 3575m).
   Note: if unable to unlatch completion string due to string pack off with huge amount of debris follow contingency plan#03
   Note: if unable to unlatch completion string due to string wash out follow contingency plan#04

Note: if observe any damage on THS follow contingency plan#05

10. RIH with 6-1/8" Mill to 3575m (Top of permanent Packer), Clear the Well CBU, POOH.
11. M/U Tandem scraper BHA to scrap 9 5/8" casing and inside 7" liner (To top of Permanent Packer), CBU, POOH 7" scraper BHA and L/D same.
12. RIH with Wire Line Corrosion Logging Tools to Top of Permanent Packer bottom, log up to surface.
Note: Based on log result and client confirmation Tie-Back operation will be done. (Extra Work).
Note: Based on Client decision & approval CBL/VDL log of 7" liner will be done.
In case of poor or free pipe Cement bonding result, appropriate side track plan and new time breakdown will be issued. (Extra Work).

13. RIH with DP, spot cement plug from 2340 to 2240 m, WOC to get hard, RIH & Tag CMT, POOH to surface.
14. RIH w/ 6 1/8" clean out BHA to TOC, D/O cement to 2265 (a minimum of 5m hard cement need t o
Be drilled out), CBU, POOH to surface and L/D 6 1/8" Drill out BHA.
Note: for test Cement plug follow contingency plan#6
Note: for dry test 7" Liner lap follow contingency plan#7

15. RIH with 9-5/8" scrapper & scrap inside 9-5/8" casing (whipstock setting depth), CBU, POOH to surface.
16. RIH RTTS PKR Inside 7" liner & Perform plugs dry test, POOH to surface.
Note: If dry test is not ok, follow contingency plan #6.

17. RIH RTTS PKR & perform 7" liner lap & plugs dry test, POOH to surface.
Note: If dry test is not ok, follow contingency plan #7.

18. Run & set 9-5/8" whipstock at 2265^{m}, open window, drill 3^{m} new formation, CBU, POOH to surface.
19. P/U & M/U New 8-3/8" Tri-cone bit and directional BHA (LWD, MWD), RIH to Btm.
20. Commence directional drilling formation from 2268^{m} to 2318^{m}.
21. Perform hole condition trip, POOH to surface.
22. Test BOP stack as NIOC Policy.
23. P/U & M/U New 8-3/8" PDC bit and directional BHA, RIH to Btm.
24. Commence directional drilling formation from 2318 to 2518^{m}.
25. CBU, Perform wiper trip, RIH back to Btm.
26. Cont. directional drilling formation to 2718^{m}.
27. Perform hole condition, POOH to surface.
28. P/U & M/U 8-3/8" PDC bit & directional BHA & RIH to Btm.
29. Continue directional drilling to 2918^{m}.
30. CBU, Perform wiper trip, RIH back to Btm.

31. Cont. directional drilling formation to 3118^{m}.
32. Perform hole condition, POOH to surface.
33. Test BOP stack as NIOC Policy.
34. P/U & M/U New 8-3/8" PDC bit and directional BHA, RIH to Btm.
35. Continue directional drilling formation to 3318^{m}.
36. CBU, Perform wiper trip, RIH back to Btm.
37. Cont. directional drilling formation to 3518^{m}.
38. Perform hole condition, POOH to surface.
39. P/U & M/U 8-3/8" PDC bit & directional BHA & RIH to Btm.
40. Continue directional drilling formation to section TD at 3728^{m}.
41. Perform hole condition prior to run logging equipment, spot pill.
42. POOH to surface.
43. Test BOP stack as NIOC Policy.
44. R/U WL/TCL and perform reservoir full suit log, then POOH.
45. P/U & M/U 8-3/8" bit & BHA, RIH & Perform condition trip for 7" liner running, CBU, POOH.
46. R/U CSG running equipment, run 7" liner to 3728^{m} (Btm).
47. Perform 7" liner cementing operation.
48. POOH with liner hanger setting tools to 3000^{m}, flush string & CBU.
49. POOH to surface, L/D liner hanger setting tools.
50. P/U & M/U 8-3/8" MT bit, RIH to 7" liner lap @+/- 2165^{m} while D/O CMT. (if any CMT: considered 7stds wet for CMT job), CBU, POOH.
51. P/U & M/U 6-1/8" MT bit & RIH to TOC, D/O POB and C/O inside liner to top of LDC, Perform CIT to 3000 psi, CBU, POOH to surface.
52. P/U & M/U 9-5/8" scrapper with same BHA & 8-3/8" rock bit, RIH to 20^{m} above the liner lap, scrap the dry test packer setting depth, POOH.
53. P/U dry test equipment & run to 20^{m} above the liner lap, perform 7" liner lap dry test as per program, secure the well, POOH to surface & L/D tools.
54. RIH with 6-1/8" bit while P/U 3-1/2" & 5" DPs as tandem scrapper BHA, Cont. C/O inside 7" liner to 5^{m} above the shoe, check & report CMT hardness before performing SBT log, CBU, POOH to surface.
55. P/U & M/U 7" CSG scrapper & perform scrapper trip inside 7" liner prior to run SBT log.
56. R/U WL logging equipment, run SBT tools inside 7" liner. POOH & L/D same.
57. RIH with 6-1/8" rock bit while P/U 3-1/2" & 5" DPs as tandem scrapper BHA, D/O 7" liner shoe to total depth (TD), displace the hole with new mud, commence drilling 5^{m} of new section, CBU, POOH to surface.
58. Perform FIT by using cement unit.
59. P/U & M/U New 6-1/8" Rock Bit and Directional BHA (LWD, MWD), RIH to Btm.
60. Continue Drilling 6-1/8" Hole Section f/3728 m to 3888.5 m.
61. Perform Hole condition POOH to surface.

Note: for Reservoir Full Suite Log & Scraper trip for 5" Liner hanger setting area follow contingency plan#8

62. R/U WL/TLC and perform reservoir Full Suit Logs, then POOH.
63. RIH w/ 7" Scraper clean out BHA and scrape around 3628m for Liner Hanger setting area, POOH.
64. Perform condition trip for 5" liner running (5" flush connection), CBU, POOH.
65. R/U Casing running gear, run 5" Liner to Btm, and hang same with 7" * 5" Liner Hanger packer @ 3628 m (100m overlap).
66. Perform 5" Liner cementing operation.
67. POOH w/ Hanger setting tool to 3000m, flush string & CBU.
68. POOH to surface & L/D Hanger setting tool.
69. P/U & M/U 6-1/8" MT Bit BHA, RIH to L.Lap @ +/- 3628m while D/Out cement (If any cement: Considered 7STD wet for CMT Job), Circ & clean the well, POOH.
70. L/D excess 5" DPs/HWDP.
71. P/U & M/U 4-3/4" DCs & 3-1/2" DPs/HWDPS, R/Back same in derrick.
72. P/U & M/U 4-1/8" MT Bit BHA, RIH to tag POB, D/Out POB, RIH to LDC depth @ +/- 3865m while D/Out any cement, work on junk sub and CBU. Perform 5" Liner CIT. (80% of Burst press) with 2000 psi, for 15 min. CBU , POOH to Surface.
Note: for dry test 5" Liner lap & Perform CMT quality Logging of 5" Liner follow contingency plan#9

73. RIH with 4-1/8" MT Bit and 7" scraper to 3865 m, CBU, POOH to Surface.
RIH with 7" RTTS & Dry test BHA, RIH to L.Lap @ +/- 3628 m, Perform 5" Liner lap Dry Test (based on Gadvan Reservoir Datum Pressure), POOH.
Note: If dry test is not ok, follow contingency plan #09.

74. Change Ram, install shooting nipple, R/U WL/TLC Logging Eq. Perform CMT quality Logging of 5" Liner from 5" liner LDC @ 3855 m to 7" Liner Lap @ 2165 m (SBT/CBL/VDL), POOH to Surface.
75. P/U & M/U TCP Assy, RIH to perforation depth, correlate depth with WL and perforate Guns as program. POOH to Surface.
76. RIH with GR-CCL TLC log to confirm perforation intervals, POOH.
77. RIH scraper and scrape 7" production packer setting area,
displace the well with Calcium Chloride (CaCl2) as completion fluid, POOH to surface.
78. RIH w/ DP's and DC's, POOH & L/D same.
79. Run 7.5K completion string as per program.
80. Install Plug & Prong inside No-Go Landing Nipple, Set BPV inside THS, N/D BOP's & install new XMT.
81. Retrieve Plug & Prong & BPV, install TWCV and Pressure test the XMT.
82. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.

83. Retrieve TWCV, R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.
84. RIH with CT to bottom, displace the well with diesel or nitrogen & observe well, POOH.
85. R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.
86. Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.
87. R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L
88. RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve.
89. Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs.
Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded).
Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.

90. RIH with S/L, set protection Sleeve, POOH & R/D SL.
91. RIH with C/T to bottom & displace the well with diesel, POOH.
92. RIH with S/L, install plug & prong into NO GO landing nipple, POOH.
93. RIH with S/L, retrieve protection sleeve.
94. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.

Contingency Plans of Well AZNS-075 Work-over:
Based on operation situation in each step, the following contingency plans might be followed:
Contingency plan #1: R/U & RIH W/Colliding Tools & Collide Completion String from Proper Depth, POOH & L/D
Contingency plan #2: Displace well with heavy mud and try to kill the well, Circulate & Balance MW.
Contingency plan #3: R/U & RIH with FPIT tools to check string free point, POOH to Surface & L/D. R/U & RIH W/Colliding Tools & Collide Completion String from Proper Depth, POOH & L/D.POOH Completion String from Colliding Depths in Several Runs.
Contingency Plan#04: Pull Completion String until string will be part from wreaked point then POOH & L/D String in Several attempt. Mill the remainder of fish & packer.
Contingency Plan#05 if observe any leakage and or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG). Note: Proper Equipment shall be provided by client and install by contractor.
Contingency plan#06: if observe any leakage and or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG). Note: Proper Equipment shall be provided by client and install by contractor.
Contingency Plan#07 RIH RTTS PKR & perform 7" liner lap dry test, POOH to surface.

Note: If dry test is not ok, P/U & M/U 7" SN Packer, RIH and set a 7" SN Packer, POOH to Surface will be followed. Stand back required DPs, HWDPs and DC.

Contingency plan#08 Perform condition trip R/U WL/TLC and perform reservoir Full Suit Logs, Then POOH. Perform Scraper Trip for 5" Liner Hanger setting area & POOH. Stand back required DPs, HWDPs and DC. Stand back required DPs, HWDPs and DC.

contingency plan#09 Perform Scraper Trip, RIH with 7" RTTS & Dry test BHA, RIH to L.Lap @ +/- 3628m, Perform 5" Liner lap Dry Test (based on Gadvan Reservoir Datum Pressure), POOH.

If dry test is not ok, P/U & M/U 5" SN Packer, RIH and set a 5" SN Packer on Top of Liner, POOH to Surface will be followed. Stand Back required DPs, HWDPs and DC.

Change Ram, Install shooting nipple, R/U WL/TLC Logging Eq.

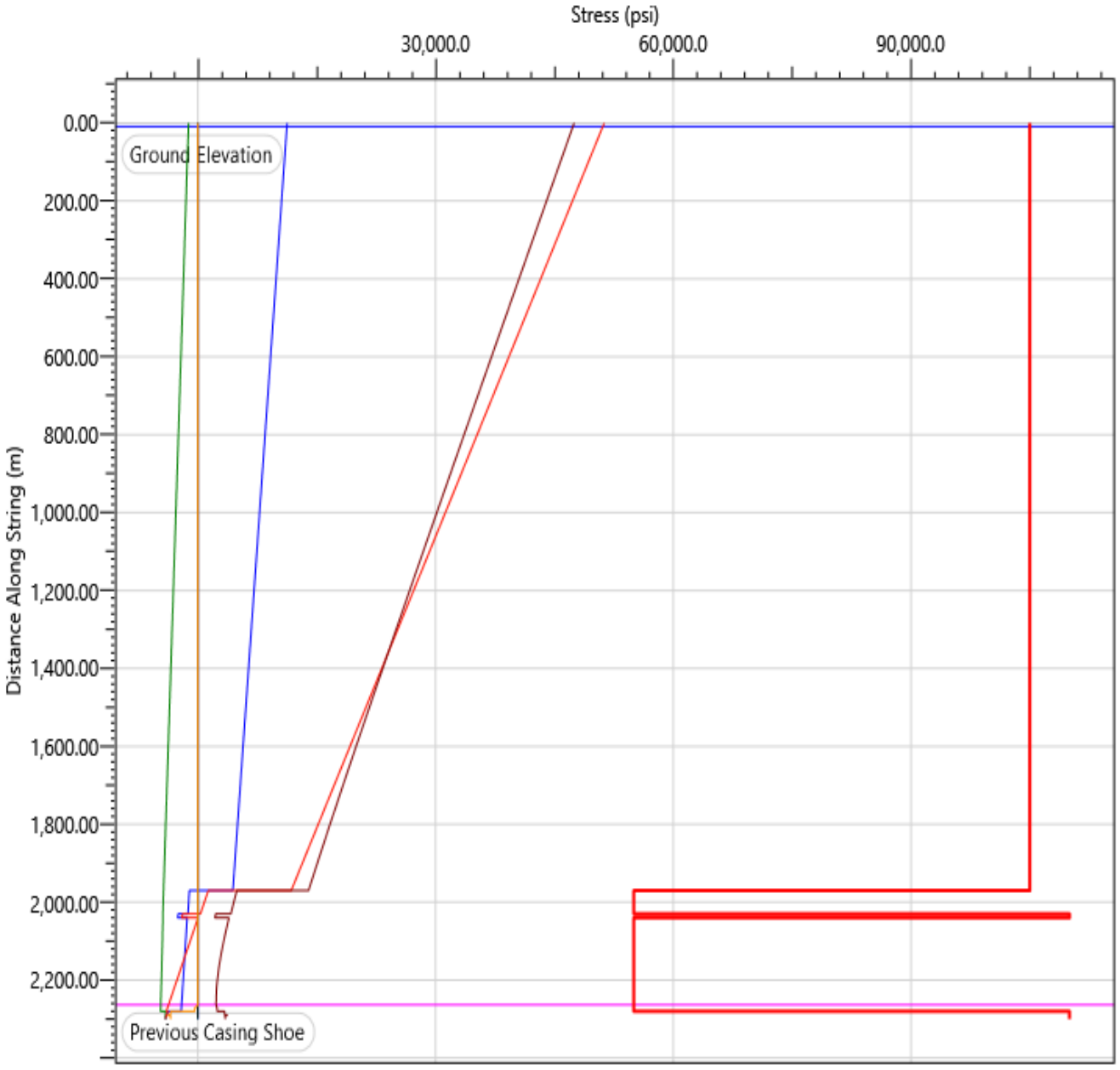
Perform CMT quality Logging of 5" Liner from 5" liner LDC @ 3855 m to 7" Liner Lap @ 2165 m (SBT/CBL/VDL), POOH to Surface.

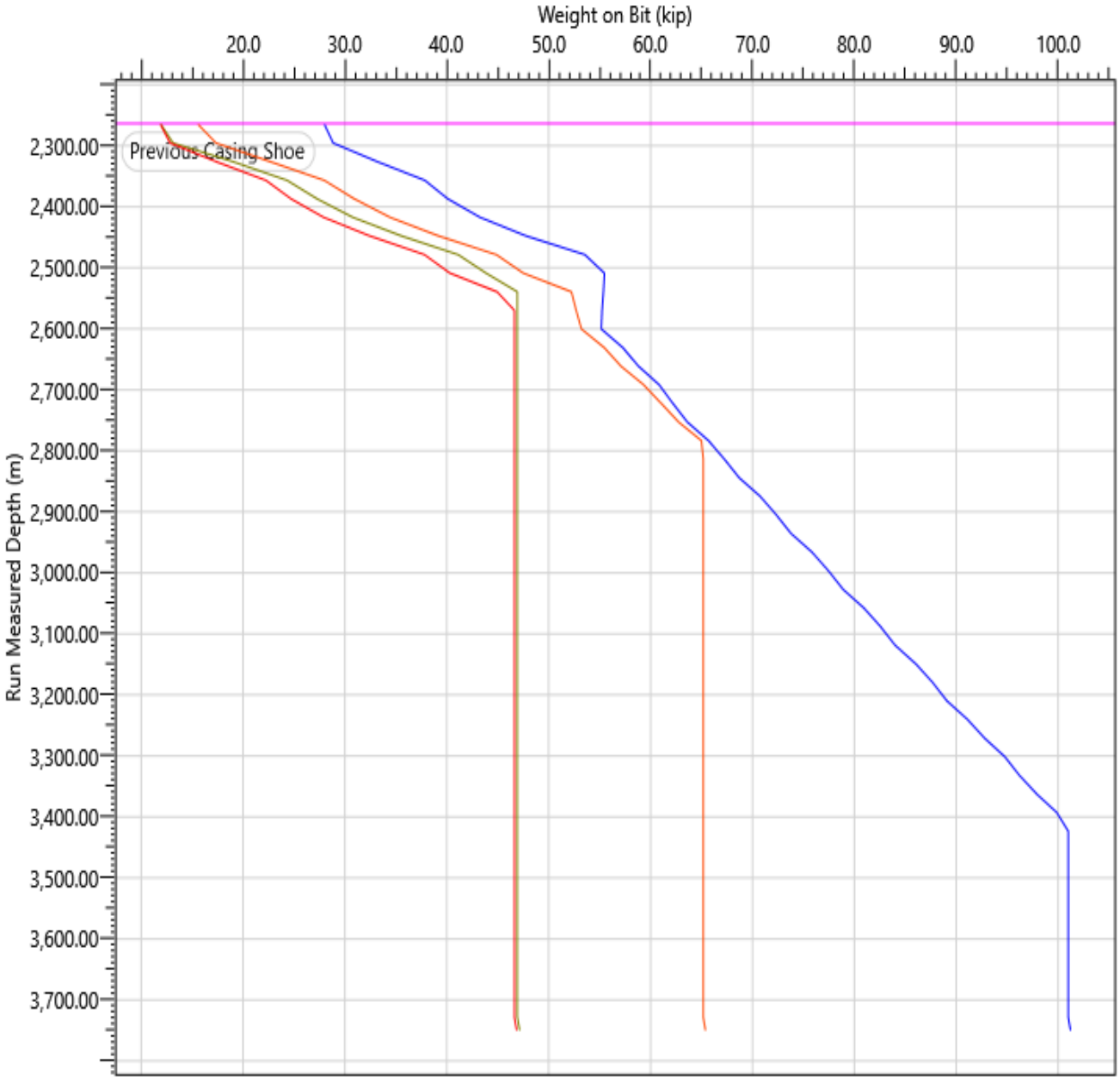
RIH with GR-CCL TLC log to confirm perforation intervals, POOH.

5. Side Track Trajectory Plan

RTE: 12.95m					GLE: 3.95m						
Vertical section was calculated toward azimuth 90deg.											
MD (m)	Inc (°)	Azi (°)	TVD (m)	TVDss (m)	N/S (m)	E/W (m)	VSEC (m)	DLS (°/30m)	Map N (m)	Map E (m)	Remarks
2265	0.72	266.01	2264.88	-2251.93	0.56	-2.81	-2.81	0	3436066.56	186424.19	Set whipstock and open window through 9 5/8" from 2265m (inside of Tarbur Mbr.)
2295	1.78	90	2294.87	-2281.92	0.54	-2.53	-2.53	2.5	3436066.54	186424.47	
2325	4.28	90	2324.83	-2311.88	0.54	-0.95	-0.95	2.5	3436066.54	186426.05	
2327.58	4.5	90	2327.40	-2314.45	0.54	-0.75	-0.75	2.5	3436066.54	186426.25	Lower Gurpi Mbr.
2355	6.78	90	2354.68	-2341.73	0.54	1.95	1.95	2.5	3436066.54	186428.95	
2385	9.28	90	2384.39	-2371.44	0.54	6.14	6.14	2.5	3436066.54	186433.14	
2415	11.78	90	2413.88	-2400.93	0.54	11.62	11.62	2.5	3436066.54	186438.62	
2445	14.28	90	2443.11	-2430.16	0.54	18.39	18.39	2.5	3436066.54	186445.39	
2475	16.78	90	2472.01	-2459.06	0.54	26.42	26.42	2.5	3436066.54	186453.42	
2476.77	16.93	90	2473.7	-2460.75	0.54	26.94	26.94	2.5	3436066.54	186453.94	Ilam Fm.
2505	19.28	90	2500.53	-2487.58	0.54	35.71	35.71	2.5	3436066.54	186462.71	
2535	21.78	90	2528.62	-2515.67	0.54	46.23	46.23	2.5	3436066.54	186473.23	
2556.35	23.56	90	2548.32	-2535.37	0.54	54.46	54.46	2.5	3436066.54	186481.46	Laffan Mbr.
2563.7	24.18	90	2555.04	-2542.09	0.54	57.43	57.43	2.5	3436066.54	186484.43	Sarvak Fm.
2565	24.28	90	2556.23	-2543.28	0.54	57.97	57.97	2.5	3436066.54	186484.97	
2595	26.78	90	2583.30	-2570.35	0.54	70.9	70.9	2.5	3436066.54	186497.9	
2625	29.28	90	2609.77	-2596.82	0.54	85	85	2.5	3436066.54	186512	
2633.6	30	90	2617.25	-2604.3	0.54	89.25	89.25	2.5	3436066.54	186516.25	M.B.U.A.P.
2655	30	90	2635.78	-2622.83	0.54	99.95	99.95	0	3436066.54	186526.95	
2685	30	90	2661.76	-2648.81	0.54	114.95	114.95	0	3436066.54	186541.95	
2715	30	90	2687.74	-2674.79	0.54	129.95	129.95	0	3436066.54	186556.95	
2745	30	90	2713.72	-2700.77	0.54	144.95	144.95	0	3436066.54	186571.95	
2775	30	90	2739.71	-2726.76	0.54	159.95	159.95	0	3436066.54	186586.95	
2805	30	90	2765.69	-2752.74	0.54	174.95	174.95	0	3436066.54	186601.95	
2835	30	90	2791.67	-2778.72	0.54	189.95	189.95	0	3436066.54	186616.95	
2865	30	90	2817.65	-2804.7	0.54	204.95	204.95	0	3436066.54	186631.95	
2895	30	90	2843.63	-2830.68	0.54	219.95	219.95	0	3436066.54	186646.95	
2925	30	90	2869.61	-2856.66	0.54	234.95	234.95	0	3436066.54	186661.95	
2955	30	90	2895.59	-2882.64	0.54	249.95	249.95	0	3436066.54	186676.95	
2985	30	90	2921.57	-2908.62	0.54	264.95	264.95	0	3436066.54	186691.95	
3015	30	90	2947.55	-2934.6	0.54	279.95	279.95	0	3436066.54	186706.95	
3045	30	90	2973.53	-2960.58	0.54	294.95	294.95	0	3436066.54	186721.95	
3075	30	90	2999.51	-2986.56	0.54	309.95	309.95	0	3436066.54	186736.95	
3105	30	90	3025.49	-3012.54	0.54	324.95	324.95	0	3436066.54	186751.95	
3135	30	90	3051.47	-3038.52	0.54	339.95	339.95	0	3436066.54	186766.95	
3165	30	90	3077.46	-3064.51	0.54	354.95	354.95	0	3436066.54	186781.95	
3195	30	90	3103.44	-3090.49	0.54	369.95	369.95	0	3436066.54	186796.95	
3225	30	90	3129.42	-3116.47	0.54	384.95	384.95	0	3436066.54	186811.95	
3255	30	90	3155.40	-3142.45	0.54	399.95	399.95	0	3436066.54	186826.95	
3285	30	90	3181.38	-3168.43	0.54	414.95	414.95	0	3436066.54	186841.95	
3308.37	30	90	3201.62	-3188.67	0.54	426.63	426.63	0	3436066.54	186853.63	Kazhdumi Fm.
3315	30	90	3207.36	-3194.41	0.54	429.95	429.95	0	3436066.54	186856.95	
3345	30	90	3233.34	-3220.39	0.54	444.95	444.95	0	3436066.54	186871.95	
3375	30	90	3259.32	-3246.37	0.54	459.95	459.95	0	3436066.54	186886.95	
3405	30	90	3285.30	-3272.35	0.54	474.95	474.95	0	3436066.54	186901.95	
3435	30	90	3311.28	-3298.33	0.54	489.95	489.95	0	3436066.54	186916.95	
3465	30	90	3337.26	-3324.31	0.54	504.95	504.95	0	3436066.54	186931.95	
3495	30	90	3363.24	-3350.29	0.54	519.95	519.95	0	3436066.54	186946.95	
3525	30	90	3389.22	-3376.27	0.54	534.95	534.95	0	3436066.54	186961.95	
3545.51	30	90	3406.99	-3394.04	0.54	545.2	545.2	0	3436066.54	186972.2	Dariyan Fm.
3555	30	90	3415.21	-3402.26	0.54	549.95	549.95	0	3436066.54	186976.95	
3585	30	90	3441.19	-3428.24	0.54	564.95	564.95	0	3436066.54	186991.95	
3615	30	90	3467.17	-3454.22	0.54	579.95	579.95	0	3436066.54	187006.95	
3645	30	90	3493.15	-3480.2	0.54	594.95	594.95	0	3436066.54	187021.95	
3675	30	90	3519.13	-3506.18	0.54	609.95	609.95	0	3436066.54	187036.95	
3705	30	90	3545.11	-3532.16	0.54	624.95	624.95	0	3436066.54	187051.95	
3728.58	30	90	3565.53	-3552.58	0.54	636.74	636.74	0	3436066.54	187063.74	Gadvan Fm. & 7" Liner Shoe @ 3729m
3735	30	90	3571.09	-3558.14	0.54	639.95	639.95	0	3436066.54	187066.95	
3765	30	90	3597.07	-3584.12	0.54	654.95	654.95	0	3436066.54	187081.95	
3795	30	90	3623.05	-3610.1	0.54	669.95	669.95	0	3436066.54	187096.95	
3825	30	90	3649.03	-3636.08	0.54	684.95	684.95	0	3436066.54	187111.95	
3855	30	90	3675.01	-3662.06	0.54	699.95	699.95	0	3436066.54	187126.95	
3885	30	90	3700.99	-3688.04	0.54	714.95	714.95	0	3436066.54	187141.95	
3888.47	30	90	3704.00	-3691.05	0.54	716.69	716.69	0	3436066.54	187143.69	T.D. & 5" Liner Shoe @ 3888m

6. Torque & Drag analysis

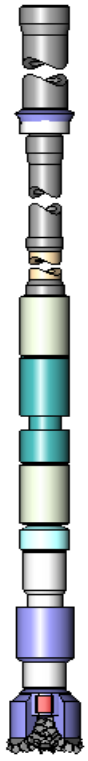



Weight on Bit (kip)
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100.0
Previous Casing Shoe
Run Measured Depth (m)
2,300.00
2,400.00
2,500.00
2,600.00
2,700.00
2,800.00
2,900.00
3,000.00
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3,600.00
3,700.00

7. BHA Program

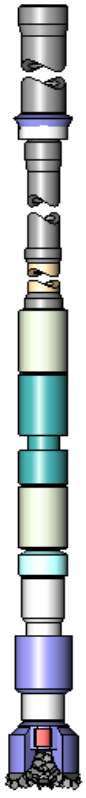
8.1 6" Milling BHA

Mill SIZE (inch)	5-7/8" or 6"	BHA Configuration to C/Out Inside 7" Liner	Collar	OD
Depth In/Out (m)			DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	85			
B.FACTOR	0.827			
Weight in Air (lb)	29,642			
Bouyed Weight (lb)	24,499			
Bouyed Weight Below Jar (lb)	17,362			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	343.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	6" Mill	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

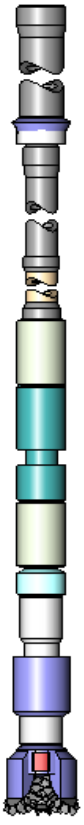
8.2 C/Out Inside 7" Liner BHA

BIT SIZE (inch)	6-1/8"	**BHA Configuration to C/Out Inside 7" Liner**				**Collar**	**OD**
Depth In/Out (m)	2356-3766					DC-1	4-3/4"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	85						
B.FACTOR	0.827						
Weight in Air (lb)	29,642						
Bouyed Weight (lb)	24,499						
Bouyed Weight Below Jar (lb)	17,362						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	345.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	344.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	229.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	174.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	147.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	139.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	2.05
	Junk Sub	1	4-1/4	1-1/2	3-1/2" Reg. B x P	1	1.3
	6-1/8" MT Bit	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

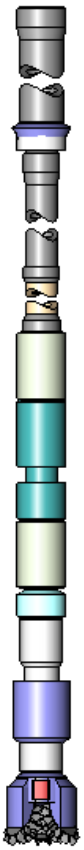
8.3 8-3/8" Directional Drilling BHA

BIT SIZE (inch)	8-3/8"	BHA Configuration to Drill 8-3/8" Hole Section (PDC Bit)				Collar	OD
Depth In/Out (m)	2265 - 3729					DC-1	4-3/4"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	76						
B.FACTOR	0.845						
Weight in Air (lb)	19,197						
Bouyed Weight (lb)	16,219						
Bouyed Weight Below Jar (lb)	15,973						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" HWDP	6	5	3	4-1/2" IF P x B	56	305.73
	6 1/2" JAR	1	6-1/2	2.25	4-1/2" IF P x B	6.39	249.73
	6 1/2" FLEX JOINT	1	6-1/2	2.25	4-1/2" IF P x B	2.4	243.34
	5" HWDP	24	5	3	4-1/2" IF P x B	222	240.94
	6 3/4" NMDC	1	6-3/8	3.5	4-1/2" IF P x B	9.22	18.94
	6 3/4" UBHO	1	6-3/4	3	4-1/2" IF P x B	1	9.72
	6 3/4" FLOAT SUB	1	6-3/4	-	4-1/2" IF P x B	0.42	8.72
	6 3/4" PDM	1	6-3/4	-	4-1/2" Reg. P x 4-1/2" IF B	8.02	8.3
	8-3/8" PDC Bit	1	8-3/8	-	4-1/2" Reg. P x B	0.28	0.28

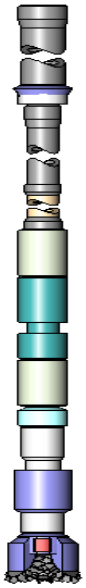
8.4 6-1/8" Directional Drilling BHA

BIT SIZE (inch)	6-1/8"	BHA Configuration to Drill 6-1/8" Hole Section (PDC Bit)				Collar	OD
Depth In/Out (m)	3728 - 3888					DC-1	4-3/4"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	85						
B.FACTOR	0.827						
Weight in Air (lb)	22,381						
Bouyed Weight (lb)	18,498						
Bouyed Weight Below Jar (lb)	16,311						

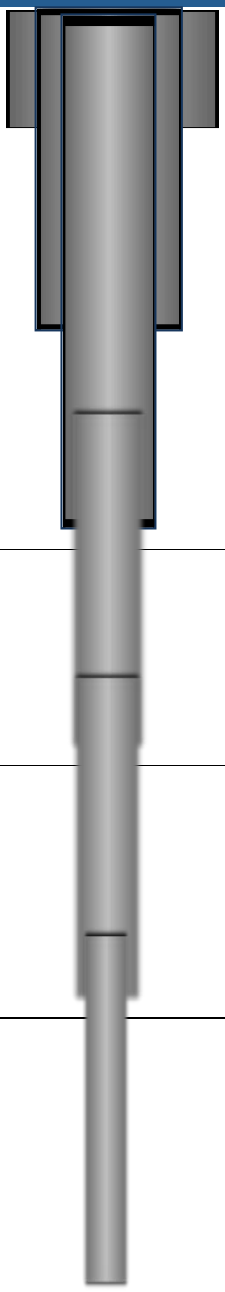
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	1764.8
	3-1/2" Drill Pipe	210	3-1/2	2-7/16	3-1/2" IF P x B	1450	1763.8
	2 Std 3 1/2" HWDP	6	3 1/2	2.25	3-1/2" IF P x B	58	313.8
	4 3/4" Drlg Jar	1	4 3/4		3-1/2" IF P x B	6	255.8
	8 Std 3 1/2" HWDP	24	3 1/2		3-1/2" IF P x B	232	249.8
	4-3/4" NMDC	1	4-3/4	2.25	3-1/2" IF P x B	9.5	17.8
	4 3/4" UBHO	1	4-3/4		3-1/2" IF P x B	1	8.3
	4 3/4" F. Sub	1	4-3/4		3-1/2" IF P x B	1	7.3
	4-3/4" PDM	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	6	6.3
	6-1/8" PDC Bit	1	6-1/8	-	3-1/2" Reg. P x B	0.3	0.3

8.5 BHA for C/Out inside 5" Liner

Mill SIZE (inch)	4"	BHA Configuration to C/Out Inside 5" Liner				Collar	OD
Depth In/Out (m)						DC-1	4-3/4"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	85						
B.FACTOR	0.827						
Weight in Air (lb)	13,634						
Bouyed Weight (lb)	11,269						
Bouyed Weight Below Jar (lb)	20,835						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	3-1/2" Drill Pipe	-	5	4.276	3-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF B x 2-7/8" HTPAC P	1	371.99
	2-7/8" Drill Pipe HT Pac	30	2-7/8	2.151	2 7/8" HTPAC	170	370.99
	3-1/8" Drill Collar	3	3-1/8	2-7/8	3-1/8" HTPAC	27.42	200.99
	3" Jar	1	3	2		8	173.57
	3-1/8" Drill Collar	18	3-1/8	2-7/8	3-1/8" HTPAC	164.52	165.57
	3-1/8" Bit Sub W/ Float Valve	1	3-1/8	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	4" Mill	1	4	-	3-1/2" Reg. P x B	0.3	0.3

8. Mud & Cement Program

Hole Size (in)	Casing Size (in)	Casing Program	Depth (m MD)	Drilling Fluid system	Mud Properties
26”	20”		151.5 m	**Already Has been Drilled**	
17 ½”	13 3/8”		1155.5 m	**Already Has been Drilled**	
12 ¼”	9 5/8”		2477.5 M	**Already Has been Drilled**	
8 1/2”	7” Liner		3799.5 M	**Already Has been Drilled**	
8 3/8” Side-Track	7” Liner		3728 M Side-Track point 2265 m	**Caco3/Polymer Mud**	Mw:72 – 85 pcf PV: ALAP YP: 12 – 18 lb/100 ft² F.L: < 3 cc/30 min PH: 9 - 10 Vis: 38 - 45 sec/qt
6 1/8”	5” Liner		3888.9 m	**Caco3/Polymer Mud**	Mw:72 – 85 pcf PV: ALAP YP: 12 – 18 lb/100 ft² F.L: < 3 cc/30 min PH: 9 - 10 Vis: 38 - 45 sec/qt

9.1 8 3/8" section (2265 – 3728^{m})

9.1.1 Challenges

- Hole instability issues
- Wellbore losses
- Shale
- Tight hole & well instability

9.1.2 Recommendations

- It is mandatory to monitor closely the rheological properties in order to ensure adequate hole cleaning, run Hydraulic calculation to ensure that mud carrying capacity is enough to meet this critical goal.
- Mud can be treated with additional shale inhibitor if deemed necessary to control shale reactivity and washouts.
- Optimize the procedure to fight the losses (see the guide at the end of this mud program) and be ready with adequate LCM material to mix LCM pill as quick as possible.
- The previous mud in hole must be dump, due to have enough surface volume, keep 300^{bbl} heavy mud in mud pits as kill mud.
- The concentration of KCl in the system is insufficient; shale cuttings will be soft and mushy. If the concentration of KCl is sufficient, they will retain their integrity. As drilling commences the potassium ion becomes depleted and must be replenished. The rate of depletion provides an indication of how reactive the shales are. While drilling gumbo, the appearance of soft mushy cuttings at surface is a good indication that the potassium concentration should be increased. Other indicators include a drilled solids increase, a CEC increase and the occurrence of tight hole or mud rings.
- Since this system is very solids-sensitive, it is best to mix the system from scratch instead of converting an existing drilling fluid (containing drill solids) to a potassium chloride system.
- This system is very sensitive to solids and hardness contamination so the first step in mixing this system is to treat the hardness.
- Due to necessary of directional drilling, it is recommended to pump (at the first 20^{bbl} Low Vis pill follow by 20^{bbl} Hi Vis Pill should be pumped every stand) for clean the hole. For better understanding of hole cleaning procedure based on hole angels please refer to technical appendix.
- Due to face with torque and drag, it is recommended to mix water base mud lubricant 1 – 3% vol. reduce torque and drag while drilling horizontal or vertical.
- In the directional drilling section, it is recommended to use Glass beads 2 – 3^{ppb} as due to have torque and drag.

- The mud weight in this interval is critical for successful drilling, hole condition will dictate the density. Mud weight will be increased by adding $CaCO_3$ F.

9.1.3 Interval Description

After drill out CMT the 8 – 3/8" section will be drilled from window point to an estimated section TD, KCL Polymer mud system necessitating a minimum requirement of 72^{pcf} to 85^{pcf} based on nearby well data.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

8 3/8" HOLE SIZE **FROM : 2265 MT**	**7" Liner Size** **TO : 3728 MT**
1463 meter to drill in Gurpi, Ilam, Lafan Sarvak, Kazhdomi, Darian, Gadvan by $CaCO_3$/Polymer mud	
DRILLING FLUID SYSTEM	**$CaCO_3$/Polymer Mud**
Key products	Limestone Powder , Salt (NaCl), Xanthan gum, Starch LV , Caustic Soda, Soda ash, KCL, Biocide
Water type	Drill Water
Solid Control	Shale shakers , Desilters , Desander, Centrifuge

Gurpi: Argillaceous limestone and marl with scattered glauconitic, asphalt, with salt aquifer. Salt water 1.15sg.

Ilam: Limestone, argillaceous limestone, shale interbed. Salt water 1.15sg.

Lafan: Shale & Dolomite.

Sarvak: Limestone to argillaceous limestone, the lower part is shale and dolomite, with salt aquifer. Salt water 1.15sg. H_2S

Kazhdomi: Shale interbedded with argillaceous limestone, sandy claystone, claystone, marl, sandy limestone, limestone beds.

Daryan: shale interbedded with argillaceous limestone.

Gadvan: Limestone, Claystone, Sandy Claystone & Sandstone.

9.1.4 Product concentration & function (before side – track)

Salt /$CaCO_3$/ Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5-1	Alkalinity Control
Soda Ash	1	Hardness Remover
Salt	Up t0 109	non - solid weighting agent
Limestone Powder	10 to 114	weighting agent
Starch LV	4 - 6	Fluid loss controller
Xanthan gum	0.5 - 1	Viscosifier
Biocide	As Requirement	Anti-bacteria

MIXING PROCEDURE

1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer
2. Fill the mud tank with Drill water
3. add Salt to reach mentioned mud weight
4. Remove Hardness by add Soda ash
5. adjust PH & alkalinty by Casutic soda
6. add Starch LV to control fluid loss
7. add Xanthan gum to reach good rheology
8. Add Limestone powder to reach Mentioned Weight
9. Check the pH and keep in range by add caustic soda
10. Mix Biocide as need

9.1.5 Product concentration & function

Salt /$CaCO_3$/ Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5-1	Alkalinity Control
Lime	0.5-1	Alkalinity Control
Soda Ash	1-2	Hardness Remover
Salt	Up t0 95	non - solid weighting agent
KCL	10.5 - 18	Shale Inhibitor
Limestone Powder	10 to 114	weighting agent
Starch LV	10-12	Fluid loss controller
Xanthan gum	0.5 - 1.5	Viscosifier
Biocide	As Requirement	Anti-bacteria

MIXING PROCEDURE

1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer
2. Fill the mud tank with Drill water
3. add Salt to reach mentioned mud weight
4. Remove Hardness by add Soda ash
5. adjust PH & alkalinty by Casutic soda
6. add Starch LV to control fluid loss
7. add Xanthan gum to reach good rheology
8. Add Limestone powder to reach Mentioned Weight
9. Check the pH and keep in range by add caustic soda
10. Mix Biocide as need

9.1.6 Required properties (before side – track)

Fluid Properties Summary		
MUD PARAMETERS		Interval
		I
Bit diameter	inch	8 3/8”
Density	pcf	72 - 85
Marsh Viscosity	sec	38 – 45
PV	cps	ALAP
Yield Point	lb/100ft²	10 – 15
Gel 10 Sec	lb/100ft²	2 - 5
Gel 10 min	lb/100ft²	4 - 7
pH	-	9 – 10
Ca++	mg/l	< 1200
Total Hardness	Mg/l	<1600
LGS (drilled Solids)	% vol	< 6
CL^-	Mg/l	120000-188000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

9.1.7 Required properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		I
Bit diameter	inch	8 3/8”
Density	pcf	72 - 85
Marsh Viscosity	sec	38 – 45
PV	cps	ALAP
Yield Point	lb/100ft²	12 – 18
Gel 10 Sec	lb/100ft²	3 - 5
Gel 10 min	lb/100ft²	4 - 8
API Filtrate	cc/30'	< 3
pH	-	9 – 10
Ca++	mg/l	< 1200
Total Hardness	Mg/l	<1600
LGS (drilled Solids)	% vol	< 6
CL^-	Mg/l	120000-188000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

9.1.8 Mud volume (before side – track)

DRILLING FLUID TYPE			
Mud Density (pcf)	72 - 85	LGS Below (%vol)	0
Bit Size (inch)	8.375	Solid Control Equipment Eff. %	0
Section TD (m)	3799	Hole Wash Out %	15
Casing OD/ID (inch)	9.625 / 8.835	Dilution Factor (bbl/m)	0
Water Depth (m)		Meters To Drilled (m)	0

Volume information (Salt Poly Mud)		
Surface System	400	bbl
Cased Hole Volume	760	bbl
New Hole Volume	0	bbl
Wash Out Volume	0	bbl
Sub Total	1,160	bbl
Dilution Volume	0	bbl
Sub Total	1,160	bbl
Carried from last section	0	bbl
Volume Received	0	bbl
Sub Total	1,160	bbl
New Vol Required	1,160	bbl
Vol transfer out	500	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1	11
Caustic Soda	55	0.5-1	21
Salt (NaCl)	2,200	Up to 109	57
Starch LV	55	4 - 6	127
Limestone Powder	2,200	up to 114	60
Xanthan Gum	55	0.5-1	21
Biocide	55	As Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	4
De foamer	De foamer	55 Gal	4
Lime	Alkalinity control/H2S anti	55 lb	80
Shell fish F,m	Loss control material	55 lb	each size 120 sxs
$CACO_3$ F,M	Loss control material	55 lb	each size 80 sxs
Mica F.M	Loss control material	55 lb	each size 120 sxs
Walnut shell F,M	Loss control material	55 lb	each size 80 sxs
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	40

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.
2-All the calculation based on max material Concentration which mentioned.

9.1.9 Mud volume

DRILLING FLUID TYPE			
Mud Density (pcf)	72 - 85	LGS Below (%vol)	6
Bit Size (inch)	8.375	Solid Control Equipment Eff. %	70
Section TD (m)	3799	Hole Wash Out %	15
Casing OD/ID (inch)	9.625 / 8.835	Dilution Factor (bbl/m)	3
Water Depth (m)		Meters To Drilled (m)	1463

Volume information (Salt Poly Mud)		
Surface System	800	bbl
Cased Hole Volume	565	bbl
New Hole Volume	327	bbl
Wash Out Volume	49	bbl
Sub Total	1,741	bbl
Dilution Volume	981	bbl
Sub Total	2,722	bbl
Excess Volume For Formation loss forecast	2,000	bbl
Volume Received	0	bbl
Sub Total	4,722	bbl
New Vol Required	4,722	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1-2	86
Caustic Soda	55	0.5-1	86
Lime	55	0.5-1	86
Salt (NaCl)	2,200	Up to 95	204
KCL	2,200	10.7 -18	39
Starch LV	55	10 - 12	1,030
Limestone Powder	2,200	up to 114	245
Xanthan Gum	55	1-1.5	129
Biocide	55	As Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	4
De foamer	De foamer	55 Gal	4
Lime	Alkalinity control/H2S anti	55 lb	80
Shell fish F,m	Loss control material	55 lb	each size 120 sxs
$CACO_3$ F,M	Loss control material	55 lb	each size 80 sxs
Mica F.M	Loss control material	55 lb	each size 120 sxs
Walnut shell F,M	Loss control material	55 lb	each size 80 sxs
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	40

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

9.2 6 – 1/8" section (3728 – 3888.5m)

9.2.1 Challenges

- Wellbore losses
- Possible HT environment
- Oil/water influx

9.2.2 Recommendations

- Hole cleaning is highly related to low shear rate parameters, so an adequate low shear rate yield point should have maintained to ensure hole cleaning. Run Hydraulic calculation to ensure that mud carrying capacity is enough to meet this critical goal.
- The previous mud will be transfer to this section.
- A suitable LCM material for reservoir should be available on location to combat down hole losses if occurred (see the guide at the end of this mud program).
- Since this system is very solids-sensitive, it is best to mix the system from scratch instead of converting an existing drilling fluid (containing drill solids) to a potassium chloride system.
- This system is very sensitive to solids and hardness contamination so the first step in mixing this system is to treat the hardness.
- Due to necessity of directional drilling, it is recommended to pump (at the first 20^{bbl} Low Vis pill follow by 20^{bbl} Hi Vis Pill should be pumped every stand) for clean the hole. For better understanding of hole cleaning procedure based on hole angels please refer to technical appendix.
- Due to face with torque and drag, it is recommended to mix WATER BASE MUD Lubricant 1 – 3% vol. reduce torque and drag while drilling horizontal or vertical
- In the directional drilling section, it is recommended to use Glass beads 2 – 3 ppb as due to have torque and drag.
- The mud weight in this interval is critical for successful drilling, hole condition will dictate the density. Mud weight will be increased by adding $CaCO_3$ F.

9.2.3 Interval description

After Drill Out CMT, the 6 – 1/8" section will be drilled from 3728 to an estimated section TD (3888.5 M) Salt Polymer mud system necessitating a minimum requirement of 72^{pcf} to 85^{pcf} based on nearby well data.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the

drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

6 1/8" HOLE SIZE FROM : 3728 MT	5" Liner Size TO : 3888.5 MT
160.5 meter to drill in Gadvan, by $CaCO_3$/Polymer mud	
DRILLING FLUID SYSTEM	**$CaCO_3$/Polymer Mud**
Key products	Limestone Powder , Salt (NaCl), Xanthan gum, Starch LV , Caustic Soda, Soda ash, Biocide
Water type	Drill Water
Solid Control	Shale shakers , Desilters , Desander, Centrifuge

Gurpi: Limestone, Claystone, Sandy Claystone & Sandstone

9.2.4 Product concentration & function

Salt /$CaCO_3$/ Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5-1	Alkalinity Control
Lime	0.5-1	Alkalinity Control
Soda Ash	1-2	Hardness Remover
Salt	Up t0 109	non - solid weighting agent
KCL	3.5 – 10.7	Shale Inhibitor
Limestone Powder	10 to 114	weighting agent
Starch LV	10-12	Fluid loss controller
Xanthan gum	0.5 - 1.5	Viscosifier
Biocide	As Requirement	Anti-bacteria

MIXING PROCEDURE

1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer
2. Fill the mud tank with Drill water
3. add Salt to reach mentioned mud weight
4. Remove Hardness by add Soda ash
5. adjust PH & alkalinty by Casutic soda
6. add Starch LV to control fluid loss
7. add Xanthan gum to reach good rheology
8. Add Limestone powder to reach Mentioned Weight
9. Check the pH and keep in range by add caustic soda
10. Mix Biocide as need

9.2.5 Required properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		I
Bit diameter	inch	6 1/8”
Density	pcf	72 - 85
Marsh Viscosity	sec	38 – 45
PV	cps	ALAP
Yield Point	lb/100ft²	12 – 18
Gel 10 Sec	lb/100ft²	3 - 5
Gel 10 min	lb/100ft²	4 - 8
API Filtrate	cc/30'	< 3
pH	-	9 – 10
Ca++	mg/l	< 1200
Total Hardness	Mg/l	<1600
LGS (drilled Solids)	% vol	< 6
CL^-	Mg/l	120000-188000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

9.3 Completion Fluid Section

The completion fluid for this section will be fresh water treated with a scavenger, inhibitor and biocide.

9.3.1 Solids – Free $CaCl_2$ fluid system

This fluid is recommended to make from fresh water or is solutions made with water and the addition of one or more electrolytes. These fluids do not contain either suspended particles or viscosifiers, which mean neither filtration control nor viscosity control, are provided. Some Brine is corrosive and when corrosion inhibitors are required, compatibility studies should be performed. Some brine is toxic and precautions should be taken to prevent contact with unprotected skin. Brine solutions can be prepared to give good inhibition of clays or shales in producing formations. In addition, common contaminants do not affect the brines.

Salt solutions were formerly considered too corrosive to be used as work-over or completion fluids or as packer fluids. The corrosion problem can be solved by adjusting the pH of the solution to above 9 (min) to 10; however, pH measurements may be less important than measuring and controlling the p-alkalinity of the filtrate; this value should be kept above 1^{cc}. Lime be used for pH control.

Adjustment of pH, or alkalinity, does not necessarily ensure the solution to be corrosion resistant. It may be necessary to use a corrosion inhibitor to achieve lower corrosion rates. In some cases, an oxygen scavenger might be used in conjunction with a filming amine inhibitor.

9.3.2 Corrosion control

Corrosion control in completion fluids is important to protect tubular goods and equipment. Corrosion can be limited by controlling the fluid pH at the neutral or alkaline level, and treating out dissolved oxygen. In severe environments such as high temperature or contamination with acid gases, the addition of filming amines can limit corrosion to acceptable levels.

The completion fluid for this section will be Local water treated with a scavenger, and inhibitor.

Pit management will be a key in the successful completion of this interval and sufficient time for planning and preparation should be allowed.

All valves should be checked for leakage beforehand and defects identified to be reported and rectified by drilling contractor.

The mud pits which are to be used to mix the completion brine, should emptied and cleaned adequately and associated lines flushed with water.

The rig pumps are to displace all clean up pills as well as the $CaCl_2$ at intended weight. Prepare spacers and wash pills as per program in the tanks. Ensure that sufficient spacer is prepared to account for any dead tanks volume.

Prepare all inhibited $CaCl_2$ before starting the displacement process. After preparing the brine should be filtered through filtration unit and store in a clean pit before displacement.

9.3.3 Filtration process

Filtration provides clean fluids that will prevent damage to the formation, the reservoir and its permeability. It does this by removal of solids and particulates from the fluid that may otherwise block the open space within the rock formation containing the hydrocarbon fluid within the production zone.

Any solids and contaminates in the completion fluid will reduce its effectiveness. An efficiently filtered and completely solid free completion fluid will add value to both the productivity and the dependability of the well in the long term.

VIRA dual pod filtration cartridge unit is recommended for filtering brines and produced water in completion, stimulation, and workover operations. The pods are connected by a manifold, providing staged filtration capability. Staged filtration utilizes cartridge filters with different ratings to maximize filtration efficiency. Cartridge filtration is designed for use with fluids containing a low percent by volume of suspended solids. Cartridge filtration is not economic when fluids are contaminated with oil or other substances.

Total Solids can be measured at the wellsite using a turbidity meter. Solids that are not added to the system to enhance the performance of a brine are considered contamination. Contaminations include formation clays, precipitates, and polymer residues, among other things. These contaminations can be filtered at the well site using diatomaceous earth, a plate and frame press, and Five-micron to ten absolute cartridges.

Note: a clear completion fluid should not be sent to the wellsite with an NTU (Nephelometric Turbidity Unit) greater than 30 or a suspended-solid Concentration greater than 40 ppm and for fluid returning from the well NTU should be less than 150.

9.3.4 Displacement process

It is recommended to circulate and condition the mud prior to displacement and to spot the open hole section to conditioned mud or a solids-free fluid loss pill. Circulate and condition the mud as needed prior to pumping the clean out pills to optimum displacement properties). After opening the circulating sub, the pumping sequence is as follows:

No.	Description	Volume BBL	Comment
1	Viscous pill	50	Clean pill
2	Continue pumping Completion Fluid until cleanliness specification has been met. Dump return at surface		

✓ **Viscous Spacer as clean pill**

NaCl Brine or Fresh Water
Xanthan Gum: Viscosifier

Mixing Procedure:

1. Fill a tank with required volume NaCl Brine or Fresh Water for pump 50^{bbl} Brine.
2. Add 3 – 4 $^{lb/bbl}$ Xanthan Gum.

✓ **Completion fluid**

CaCl2 Brine at 85^{pcf}.
Raise pH to abive 10 with lime at the end and before pump to the well.

Corrosion: corrosion inhibiter packer fluid design.
H_2S Scavenger: oxygen scavenger (to eliminate dissolved oxygen as source of corrosion)

Mixing Procedure for 1^{bbl} 85^{pcf} Brine:

1. Fill the tank with 0.9^{bbl} fresh water.
2. Add 195^{ppb} $CaCl_2$.
3. Raise pH to above 10 with 3 – 4^{ppb} lime.
4. Add 1% V/V corrosion inhibitor.
5. Add 0.5% V/V H_2S scavenger.
6. Add 0.5% V/V O_2 scavenger.

RIG CLEAN UP CHECK LIST (Should be done by Rig Operator)

A. Areas to clean:

1. Surface pits.
2. Clean the suction, return, settling & reserve pits.

3. Clean the sand traps.
4. Clean the ditches & troughs.
5. Clean the trip tank.
6. Clean the shaker area including all solids equipment & lines.
7. Clean the mixing hoppers.
8. Clean the cement unit.
9. Clean any other pits or equipment that completion fluid may contact.
10. Cycle all valves in mixing system.

B. Items to flush out:

1. All mud pumps & suction lines.
2. All centrifuges & lines.
3. Chemical & weighting mixing lines in contact with the completion fluid.
4. Lines from the pit room to the trip tank.
5. Lines from the rig floor to the trip tank.
6. Choke, choke manifolds & kill line.
7. Standpipes.
8. Degassing & solids control equipment.
9. Lines to the cement unit and/or other pumping equipment.
10. Casing fill-up lines.
11. Lines to/from the filtration line.
12. Take on & return lines to & from the boat.
13. Any other lines that the completion fluid may contact.

9.3.5 Displacement procedure

Pre – Displacement Steps:

1. Run bit and dual or tandem scrapers.
2. Circulate and condition mud. Reduce funnel viscosity as much as possible while maintaining adequate rheological properties to clean hole and prevent settling of solids. Maintain critical annular velocity to assure that liner tops are thoroughly cleaned prior to displacement.
3. Rig up equipment for displacement of completion fluid. All tanks, circulating lines, and pumps to be used in the displacement should be cleared of residual mud.
4. Insure sufficient space is available to store displaced mud.

DRILLING FLUID TYPE					
Mud Density (pcf)	85	-	85	LGS Below (%vol)	
Bit Size (inch)		0		Solid Control Equipment Eff. %	
Section TD (m)		4200		Hole Wash Out %	
Casing OD/ID (inch)				Dilution Factor (bbl/m)	
Water Depth (m)				Meters To Drilled (m)	

Volume information (Cacl2 Brine)		
Surface System	250	bbl
Cased Hole Volume	750	bbl
New Hole Volume	0	bbl
Wash Out Volume	0	bbl
Sub Total	1,000	bbl
Dilution Volume	0	bbl
Sub Total	1,000	bbl
30 % Extera Vol	300	bbl
Volume Received	0	bbl
Sub Total	1,300	bbl
New Vol Required	1,300	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Lime	55	3-4	95
CaCl2	2,200	up to 195	99
Corrosion Inhibitor	55 Gal	1% V/V	10
H_2S Scavenger	55 Gal	0.5% V/V	5
O_2 Scavenger	55 Gal	0.5% V/V	5

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

9.4 Technical Appendix

9.4.1 Losses control procedure

Loss of fluid returns will lower the hydrostatic head of the drilling fluid in the wellbore, thereby inducing a kick. The influx fluid will then flow to the surface or into the zone of lower pressure. Lost circulation can occur in naturally occurring fractured, vuggy, cavernous, sub-normally pressured or pressure depleted formations. Induced losses can occur from mechanical fracturing due to pressure surges while breaking circulation. In all cases of lost circulation, attempts should be made to keep the hole full. The hole can be filled with either light drilling fluid or water. A record of the amount of fluid pumped should be made.

The concentration of LCM will vary according to losses. In the event of seepage losses, treat system with a finer grade of LCM at 20 – 30 Kg/m³. In the event of moderate to high rate of well bore loss, prepare pills at different graded concentrations and spot over thief zone. The concentration may vary between 18 to 70 lb/bbl. In order to avoid LCM plugging MWD tools a circulation sub should be installed in the drill string. The MWD engineer is to advise on the size and concentration of LCM that can be pumped through MWD Tools. Ensure that the drilling displace the LCM pill out of the string prior to shutting down the mud pumps.

Important Note: Please note below concentration and recommended materials may be change with same materials on rig site.

In case of having circulation losses:

Losses less than 10 bbl./hr.:

- Active system to be loaded with 10 lb./bbl. of Fiber Fine.

Losses 10- 40 bbl./hr.:

- Active mud to be loaded with 15- 20 lb./bbl. of Fiber Medium.
- Sweep the hole frequently with 50^{bbl} high viscous pill contains;
- 15 lb./bbl. Mica Medium - Corse.
- 10 lb./bbl. Walnut Shells Medium -Fine.
- 10 lb./bbl. Oyster Shell Fine.

Losses 40 – 100 bbl./hr.:

- Stop Drilling.
- Pull out string above the loss zone
- Spot 100^{bbl} pill contains;

- 30 lb./bbl. Fiber coarse.
- 30 lb./bbl. Walnut Fine.
- 20 lb./bbl. Mica Med.
- Repeat the previous pill minimum of two times.

Losses more than 100^{bbl}:
Up to company procedure, may be squeeze 50^{bbl} of Cement squeeze.

9.4.2 Mixing LCM Pills

The amount of LCM that can be added to a mud depends very much on the mud properties. High gels and YP will help keep the material in suspension. Since LCM is generally of low gradient, excess material tends to float on top if too much is added. The following procedure may prove useful for mixing LCM pills and avoiding the problem of flotation.

1. ENSURE the pill has enough viscosity to keep the low gradient material in suspension before addition of LCM. ADD appropriate viscosifier if required.
2. ADD the material in stages e g. 5, 10, 15, 20 etc. ppb.
3. STOP after desired concentration has been reached, or when it is observed that the material cannot be held in suspension.
4. If a higher LCM content is desired, LEAVE the pill to soak for a period of time and then TRY ADDING more material.

9.4.3 LCM Pill spotting procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30minutes and monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close the BOP and squeeze 10^{bbl} of LCM pill into the formation without exceeding 200^{psi}.
3. If the hole remains stable, continue to drill ahead.
4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix and spot another 50^{bbl} of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20^{bbl} of LCM pill into the formation without exceeding 200^{psi}. If a squeeze pressure develops, hold the pressure for 1 hour before staging the drill string back to bottom.

Notes:

- All time hole must be full, transfer mud to trip tank and asked driller to fill hole from trip tank.
- All LCM pills must be thoroughly mixed and homogenised before being pumped.

- When pumping an LCM pill through a bit with nozzle smaller than 18/64, DO NOT USED LCM Medium and Coarse sizes.
- Remove filter screens from the drill pipe and if necessary from the mud pump suction lines.
- This is particularly true if the mud density is high, 14 lb/gal or greater. Fine loss circulation material which does not adsorb a lot of water should be used in weighted mud.

GUIDELINE FOR FIELD ENGINEERS FOR SHALE INHIBITION

Shale Hydration takes two forms in water-sensitive shales: surface and osmotic adsorption. Surface hydration occurs when a small volume of water is strongly adsorbed onto the planar surfaces of the clays which causes little softening or swelling, but can lead to excessive stresses if the swelling is confined.

Osmotic swelling occurs when a large volume of weakly held water is attracted to the clay surfaces by electrostatic forces. Osmotic swelling causes softening and significant swelling as the adjacent clay layers' hydrate water and expand. Osmostic swelling does not generate excessive stresses, even when confined, and can be substantially reduced if Extremely difficult drilling conditions have led to more systematic approaches to analysing wellbore instabilities.

One such approach that can be used as a guideline for the mud engineer involves three basic steps:

1. Monitor and analyze the fundamental failure mechanism.
2. Determine and understand the factors governing this failure mechanism.
3. Model the problem considering impact of proposed changes.

9.4.3.1 MONITOR AND ANALYSIS THE FUNDAMENTAL FAILURE MECHANISM.

Monitoring of data and events on the rig is the first step of the process and involves:

- **Collection, organization and processing of drilling data**. Drilling data such as Rate of Penetration (ROP), Revolutions per Minute (RPM), Weight-On-Bit (WOB), Bottom-Hole Assembly (BHA), torque, over pull Trip in Hole (TIH) and Pull Out of Hole (POOH), swab and surge, flow rate, mud weight, mud Rehology, etc. should be collected, organized and analyzed on a continual basis and compared to the symptoms of wellbore stability being observed. This will help evaluate performance and establish trends or any deviations and measure the effectiveness of treatments and changes.

- **Formation characterization.** The standard procedure in drilling operations is to apply the knowledge gained from previous wells to subsequent wells drilled in similar geologic regions. Wellbore-instability investigation requires evaluation of shales during the drilling process. Since

rig site shale testing is limited due to equipment and time limitations, most evaluations are performed in a properly equipped laboratory.

Laboratory tests are conducted to evaluate shale swelling, dispersibility, cation exchange capacity, hydration capacity, colloidal **content**, capillary suction time, shale hardness tests and mud compatibility properties.

A number of methods have been used to perform rig site tests. These include tests as simple as the methylene blue test (CEC) measurement of shale; shale hardness measurements; hot-rolled dispersion testing with various mud formulations and additives; and sophisticated analytical measurements of various clay properties. It is particularly important to try and determine the failure mechanism: swelling and softening or brittle spalling. Information from these tests, current shaker observations and mud logger data are used to evaluate drilling fluid performance, adjust mechanical drilling parameters and making casing design decisions. The primary options to counter shale problems at the rig site are:

- Change the drilling fluid density.
- Change to an oil- or synthetic-base system.
- Change to a more inhibited water-base mud.
- Reduce fluid loss to very low levels and increase filtrate viscosity.
- Utilize pore-plugging additives, emulsified insoluble materials or plastering agents.
- Adjust mechanical drilling parameters.
- **Evaluation and analysis of instability symptoms.**

Instability warning signs such as the presence of excessive cavings on the shakers for a given ROP, fill on connections, tripping difficulties, torque, stuck pipe, etc. indicate that something is wrong down hole. The magnitude or volume of failed rock should be estimated and recorded with the other drilling parameters. The number and location of specific incidents should be recorded, such as pipe

sticking, fill or packing-off, reaming and re-drilling. All of this information can then be organized and correlated with time and depth to detect the location and quantify the severity of critical sections in the well.

9.4.3.2 DETERMINE & UNDERSTAND FAILURE MACHANISM

All of the monitored and plotted parameters are then evaluated to identify the most probable mechanism causing the instability. The most useful symptoms used to indicate the mechanism responsible for wellbore instability are:

- Caving’s analysis.

- Tripping difficulties.
- Drilling conditions.
- Mud system analysis.
- In-situ conditions.

These symptoms should be evaluated based on an interdisciplinary knowledge of the causes of wellbore instability to determine the most likely failure mechanism.

9.4.3.3 MODEL & CONSIDERING PROPOSED CHANGES

Once the monitoring and determination steps have produced parameters, organize and analyse this data to obtain qualitative solutions and evaluate proposed changes. One good way to do this is through the use of a decision tree or flowchart approach to evaluate the possible failure mechanism and to identify the most promising remedies. After arriving at a proposed solution, a corrective measure should be applied and the results used as feedback into the method and decision tree method.

A decision tree or flowchart, such as shown in Figure 9 for evaluating the cause of excessive caving's, provides a logical path for identification of the mechanism and a qualitative solution. Maintaining hole stability while drilling shale sections can be particularly troublesome. Simple solutions do not always exist, but good drilling practices combined with good mud practices are most often successful. Analysis of the mechanical stress/strain relationships has helped to develop the proper density requirements for hole stability.

9.5 Directional Drilling

9.5.1 Drilling fluid properties & selection

It is common knowledge among drilling engineers and drillers that borehole problems can be considerably more severe in a directional hole than in a vertical well. The problems are magnified as the hole angle and horizontal displacement increases. Although most of these reactions are mechanical in nature, much can be done to alleviate them by proper selection and maintenance of the drilling fluid. For example, if a water-base fluid is being used, lubricants can be added to lower the coefficient of friction between the drill string and borehole or casing. Inhibiting salts can be added to reduce the interaction of the drilling fluid with the exposed formations, thereby improving borehole stability.

There are a number of drilling factors influenced by the drilling fluid, but the following are of primary concern in drilling a horizontal well.

- Borehole instability
- Hole cleaning
- Torque and drag
- Formation damage
- Barite sag
- Hydraulics

9.5.2 Borehole instability

Borehole instability is one of the costlier, but relatively common problem associated with drilling oil and gas wells. Not only can it impede drilling, it can result in loss of the hole. The term normally conjures thoughts of borehole collapse with accompanying stuck pipe. This is one of the more critical reactions of borehole instability, but it can manifest itself in subtler ways such as hole-fill, bridges, hole enlargement, or hole size reduction.

These reactions lead to other associated problems including increased torque and drag, logging difficulties, poor cement jobs, and increased drilling days, all of which add significantly to the drilling cost. It should be pointed out that the discussions to follow do not cover hole enlargement problems from weakly consolidated sands, or rubble zones that erode from the hydraulic action of the drilling fluid.

9.5.3 Stabilization

The swelling activity is controlled by adjusting the electrolyte content of the fluid to minimize expansion of the clays. Lime and gypsum-treated fluids have long been used for this purpose, but basically do not satisfy the inhibition needs of the more troublesome shales. The potassium ion is an effective swelling

inhibitor and is used extensively for this purpose through the addition of potassium salts. The concentration of potassium utilized ranges from several hundred mg/L to over 60,000 mg/L. The most cost-effective potassium concentration, however, appears to be in the 3000 to 7000 mg/L range. In marine environments, seawater or seawater with additional sodium chloride, are good alternates to potassium. The tendency for the shale to disperse or subdivide into discrete particles appears to be best controlled by the addition of an adsorptive, high molecular weight water-soluble polymer.

One of the more effective polymers for this purpose is the partially hydrolysed poly acrylamide with molecular weight in excess of 15 million. These anionic polymers adsorb on to the surface of the cuttings and borehole wall to act as a binder or elastic coating to retard dispersion. Another approach to stabilizing the hard brittle shales that contain natural or induced micro fractures is the use of water-dispersible asphalt or Asphaltene. The addition of these type materials appear to plug the micro-fractures, or perhaps even surface pore throats to retard further fluid invasion. This action has proven very effective in certain shales and can be applied in most water base fluids.

9.5.4 Hole cleaning

Cleaning cuttings from a vertical hole has been covered extensively by numerous investigators. Hole cleaning in a vertical hole is essentially a matter of overcoming the slip velocity of the cuttings with sufficient annular velocity and carrying capacity of the drilling fluid. Although any velocity greater than the settling velocity of the largest cutting will theoretically lift all the cuttings to surface eventually, too low an annular velocity will lead to an undesirable high concentration of cuttings in the annulus. Experience has shown that cuttings concentrations more than 5% by volume cause tight hole or stuck pipe when circulation is stopped.

9.5.5 Cutting bed formation

Radial slip velocity increases with increasing angle of inclination, forcing more cuttings to the low side of the annulus, which eventually results in the formation of a cuttings bed. reported that bed formation was dramatic in the 40° to 50° angle with the bed sliding downward against the flow, resulting in a very high cuttings concentration.

The fact that a cutting bed forms is a serious problem, but the instability of the bed in this critical angle of the well makes the problem even more severe. High torque and drag resulting from the high cutting concentration at the lower part of the angle can obviously result in stuck pipe or loss of circulation.

9.5.6 Hole cleaning sweeps

In static drill string concluded that high-density sweeps were more effective than high-density high-viscosity sweeps based on field measurements. The higher viscosity of the latter caused flow diversion to the high side of the wellbore. Not surprisingly, high-viscosity sweeps also performed poorly. Also

concluded that the best hole cleaning sweeps were high-density low viscosity sweeps, with a density 3 – 4^{ppg} higher than that of the drilling fluid. the following procedure is recommended:

- Calculate sweep height in the vertical part of the annulus, and added hydrostatic head in the annulus.
- Rotate at 80+ rpm while pumping the sweep, 180+ rpm is preferred. It is not recommended to pump a weighted sweep while sliding.
- Keep pumping until the sweep is out of the hole. If pumping stops while the sweep is in the deviated hole, it falls out of the "jet stream" and re-deposits captured fines.
- Avoid increasing the viscosity of the weighted sweep.
- Plan to run 3 – 4 sweeps to achieve desired results when hole cleaning problems are indicated. One sweep will probably not be sufficient to remedy an existing situation.
- Avoid checking for flow or closing the annulus with the weighted sweep in the drill pipe.
- If desired for preventative maintenance: 1) schedule sweeps every 6 – 8 hours or 500 – 1000 feet (whichever comes first, unless the hole dictates greater frequency); barite/silt build- ups are dependent on time and footage: and 2) pump a sweep while circulating bottom up before tripping, and another once back on bottom. concluded that high-density sweeps were more effective than high-viscosity sweeps. They strongly advocated the use of sweep reports, which should be standard practice in any drilling fluids operation.

9.5.7 Short tripping

Another technique that has proven helpful in controlling problems created by the cuttings bed in the high-angle portion of the well is to make periodic short trips. Tripping the bit through the bed not only assures that the bottom hole assembly and bit can be pulled through it, but the action should agitate the bed to the point that a portion will slide down the hole and subsequently be picked up when circulation is commenced. The time intervals for these short trips will be determined by the torque and drag conditions that exist in any particular well. Lacking specific well history, a general rule of thumb is to short trip at least once for every 500 feet of new hole.

9.5.8 Drill string rotation

The industry is in total agreement that drill string rotation improves hole cleaning, especially in hole sizes less than 17 ½". The general opinion in the industry is that the optimum drill string rotation rate is approximately 120 rpm. It is believed that the drill string rotation improves hole cleaning by mechanically disturbing the cuttings bed on the low side of the hole. There could also be effects due to skewing the velocity profile in the wellbore and the creation of Taylor Vortices. Encouraged the use of rotary steerable systems.

9.5.9 Turbulence

Bottom hole assembly design becomes important because the flow restrictions of some of the components, such as the measurement-while-drilling tools or positive displacement motors, may not allow sufficient flow to reach turbulence in the annulus. Turbulence can, of course, be induced at some constant flow rate by decreasing the annular area. This may be more difficult to accomplish in a horizontal well because of limitations on the design of drill strings and tool assemblies. An increase in the effective viscosity of the drilling fluid will decrease the Reynolds Number, and thus increase the flow rate required to obtain turbulent flow. Thus, a lower effective viscosity fluid may be desirable in the horizontal well, but attention must also be given to cleaning the larger diameter portion of the upper hole or possible washed-out sections where it is impractical to maintain turbulent flow.

It is interesting to point out a basic fact between static fluid pressure and dynamic fluid pressure. Static fluid pressure is a function of true vertical depth, while dynamic fluid pressure, or ECD, is a function of measured depth. Thus, the ECD will continue to increase, albeit slowly, even after the hole becomes essentially horizontal. The fracture pressure could be reached if the hole was extended to sufficient depth and the fluid density is close to the fracture gradient.

9.5.10 Hole cleaning guide line

The single most important factor related to hole cleaning in deviated wells is flow rate. During directional drilling operations drilled cuttings will settle on the low side of the hole and form a stationary bed if insufficient annular fluid velocity is used. The critical flow rate required to prevent cuttings bed formation must be determined.

Typically, few hole cleaning problems exist in vertical or horizontal sections. Most problems associated with hole cleaning are seen on deviated wells and occur in the 40 – 60-degree section where gravity effects can cause cuttings beds to slump down the hole. Typical flow rates to aim for in ERD wells:

Hole Size	Typical Flow Rates
17 ½”	800 gpm minimum. Some rigs achieve 850 – 1000 gpm.
12 ¼”	Aim for 800 gpm (although 800 – 1000 gpm is typically achieved). If 800 gpm is not achievable, ensure tripping procedures are in place for poorly cleaned hole.
8 ½”	Aim for 400 gpm.

When correctly designed, both laminar and turbulent flow regimes will effectively clean a deviated well. In general, increasing the viscosity of a fluid in laminar flow will improve hole cleaning as will a

reduction of the viscosity of a fluid in turbulent flow. It is important that one or other regime is selected and that the transition zone between the two is avoided.

Generally viscous fluids in laminar flow are preferred because:

- It is possible to achieve higher cleaning capacity.
- Viscous fluids give better transport in the near-vertical sections.
- Viscous fluids have better suspension characteristics when circulation is stopped.
- It is difficult to achieve “turbulent flow” except in small hole sizes.

Experience has shown that good drilling fluid rheological properties are extremely important to hole cleaning when drilling a high angle well. Studies show that the effects of drilling fluid

rheological properties and annular flow regime are mutually dependent. In the laminar flow regime, increasing fluid viscosity will improve hole cleaning and this is particularly effective if the Yield Stress is high. In the turbulent regime, however, reducing fluid viscosity will help removing cuttings. Therefore, the drilling fluid rheological properties should be designed to avoid the transitional flow regime. For hole sizes above 8 ½”, the annular flow is laminar under most circumstances. Therefore, it is necessary to specify a minimum Yield Stress. The optimum level required is best established based on field data and experience.

A higher ROP requires a higher flow rate to clean the hole. It is good practice to drill the hole with a steady ROP and select the required flow rate for hole cleaning accordingly. Before making a connection, the hole should be circulated at the normal flow rate to clear the cuttings from around the BHA. Depending upon the hole angle and length of BHA, a circulation time of 5 to 10 minutes is often required.

Before tripping out, the hole should be circulated at the normal flow rate until the shakers are clean, and the drill pipe be rotated at maximum speed/reciprocated in the meantime. This may require up to 3 bottoms-ups, depending on the hole angle and hole size. Below are listed the recommended number of bottoms-up prior to tripping.

Hole Angle, degree	8 ½"	12 ¼"	17 ½"
0 – 10	1.3	1.3	1.5
10 – 30	1.4	1.4	1.7
30 – 60	1.6	1.8	2.5
60+	1.7	2.0	3.0

9.5.11 Torque & Drag

Overall friction can become so great in an extended-reach well or horizontal section that the drill string can only be rotated with great difficulty. Needless to say, this condition is conducive to stuck pipe. Obviously, this coefficient of friction will be significantly affected by the lubricating properties of the drilling fluid.

Many of the causes of torque and drag are mechanical, while others are related to hole geometry. Several, however, can be directly attributed to the drilling fluid. Since wall contact with the drill string will be extremely high in a horizontal hole, it is imperative that the lubricating characteristics of the fluid be considered.

9.5.12 Formation damage

The obvious reason for drilling a horizontal well is to expose more of the producing zone to the wellbore for increased production. Thus, the importance of preventing formation damage during drilling and completion of the horizontal well is just as important as it is in its vertical counterpart.

In fact, it may be more important since the increased cost of drilling the horizontal well must show additional production for its justification. A 50% damage or permeability loss in a potential 500 bbl/day horizontal well represents a considerably greater loss in production than a 50% damage in a 100 bbl/day vertical hole through the same reservoir.

Sandstone production is by far the most susceptible reservoir rock to damage or permeability loss during drilling and completion operations. Though formation damage may be caused by poor drilling and completion practices, one of the primary causes is due to exposure to the drilling and completion fluid.

This loss, attributable to the fluid, can be caused by one or more of the following occurrences:

- Particle transport from the drilling fluid into the production zone, subsequently plugging pore throats.

- Fluid filtrate reacting with expandable clays in the rock to decrease pore throat diameters.
- Particle movement within the permeable rock due to dispersion of clays and other minerals from quartz surfaces.
- Wettability changes of the formation from exposure to the drilling fluid filtrate.
- Interaction of fluid filtrate with formation fluids to form water - insoluble precipitants.

The effect of these actions on the permeability of sandstones has been the subject of a considerable investigation. In almost all cases, some damage is caused by the drilling and completion fluid. The primary objective is to limit this damage as much as possible. Consequently, selection of the drilling and completion fluid becomes an extremely important function in drilling the horizontal well.

9.5.13 Recommended fluid parameters

- Velocity – One of the major obstacles to overcome in horizontal and ERD drilling is hole cleaning. In vertical holes, cuttings have room to settle without causing problems. In horizontal wells, cuttings only have to settle 4+ inches to form a cuttings bed. Research and practical experience has shown that cuttings bed formation is analogous to sand dunes on an ocean beach. The bed moves in a wave fashion up the hole. If sufficient annular velocity is not present, the bed thickens until it reaches equilibrium, which usually occurs when turbulent flow conditions exist. With water, turbulent flow is easily obtained since viscosity is in the denominator of the turbulent flow equation, and water has a viscosity of 1. Fluids other than water require more pump output to approach turbulent flow conditions. Practical experience in other industries has proven that materials can be transported in water slurries if the fluid velocity is between 3 and 7 ft./sec.

- Rheological Properties – Ideally one would want a true thixotropic fluid to drill horizontally. In effect, two different conditions exist simultaneously. One occurs in the vertical portion of the hole where larger diameter casing may be set, or wash-outs in the open hole occur. In that case, a fluid having a high Yield Stress, or if the fluid is visco-elastic, the initial gel strength needs to be elevated. The opposite is needed in the horizontal section from a hole-cleaning point of view. Here one wants water for the reasons pointed out earlier. Compromises are the result. If casing is set into the zone of interest, or if formations up-hole are not a problem, then water may be possibly combined with occasional sweeps.

- Surface Apparatus – Special attention must be placed on the solids removal equipment at the surface. Solids removal efficiency must be continually monitored so that unwanted particulate material is not retained in the fluid system that may negatively affect fluid weight or filter cake quality, resulting in lost circulation or increased torque and drag. The cost of problems that may occur due to not doing this correctly may far outweigh the cost of equipment needed.

9.6 Hydrogen Sulphide Gas in Drilling Fluid

9.6.1 Description of Hydrogen Solphide Gas

Hydrogen sulphide is a colourless gas, with an offensive odour and a sweetish taste. It is soluble in water, alcohol, oils, and many other solvents. It has a specific gravity of 1.1895 with reference to air. It is considered a weak acid, it is toxic to humans and corrosive to metals. Hydrogen sulphide can be dangerous to personnel on the surface as it is extremely toxic to human and even animal life, and is extremely corrosive to most metals as it can cause cracking of drill pipe and tubular goods, and destruction of testing tools and wire lines. When hydrogen sulphide enters the borehole during drilling, completion, or testing for hydrocarbons, it creates several very detrimental problems. These problems are encountered regardless of the source of the hydrogen sulphide. Hydrogen sulphide has created unsolved problems in the oil field.

9.6.2 Origin of Hydrogen Solphide

During oil and gas operations, hydrogen sulphide can be found as a component of formation gases, dissolved in water, hydrocarbons, or even liquid sulphur. Thermal degradation of organic materials and sulphate reducing bacteria (SRB) can create hydrogen sulphide along with other gases.

9.6.3 Problems caused by Hydrogen Solphide Gas

9.6.3.1 HEALTH & SAFETY OF THE PERSONNEL

One of the main problems hydrogen sulphide causes is in respect to the health and safety and safety of the personnel. A maximum of eight hours' exposure to concentrations greater than 100 ppm H2S will cause hemorrhage and death. Concentrations above 600 ppm can be fatal in three to five minutes. Highly stressed, high strength steel can fail in a matter of minutes in the presence of 50 ppm H2S. At high pressure, as little as 0.1 ppm H2S can greatly reduce the time to failure of highly stressed, high strength steel. Relying solely on its odour is not a good idea because at concentrations above 100 ppm it deadens a person's sense of smell within a few minutes. The pure gas is heavier than air and can collect in low areas such as pit rooms and accommodation. below Table shows the potential hazards hydrogen sulphide can cause to the personnel from short-term exposures. There are many historical cases to document the ill effects of hydrogen sulphide exposure.

The documented cases reported in Table 2 indicate that hydrogen sulphide is very toxic and it takes only a matter of minutes to affect unprotected workers.

Health effects from short-term exposure to hydrogen sulphide	
Concentration (ppm)	**Health Effect**
0.01 – 0.3	Odour threshold
1 – 20	Offensive odour, possible nausea, tearing of the eyes or headaches with prolonged exposure
20 – 50	Nose, throat and long irritation; digestive upset and loss of appetite; sense of smell starts to become fatigued; acute conjunctivitis may occur (pain, tearing and light sensitivity)
100 – 200	Severe nose, throat and long irritation; ability to smell odour completely disappears.
500	Severe lung irritation, excitement, headache, dizziness, staggering, sudden collapse knockdown), unconsciousness and death within a few hours, loss of memory for the period of exposure (results in permanent brain damage if not rescued immediately).
500 – 1000	Respiratory paralysis, irregular heartbeat, collapse and death without rescue.
> 1000	Rapid collapse and death

9.6.3.2 EFFECTS OF HYDROGEN SULPHIDE ON METAL

Hydrogen sulphide mainly cause corrosion problems to drill strings, transport pipes, storage tanks etc. Hydrogen sulphide causes sulphide stress cracking, hydrogen embrittlement and pitting corrosion in oil and gas operations. The corrosion of iron in the presence of hydrogen sulphide and water is dependent upon the dissociation of the hydrogen-sulphide molecule. Iron is oxidized to the ferrous form at the anode and hydrogen sulphide under- goes a two-step dissociation at the cathode. Tung et al22propose the following equilibrium reactions below.

At the anode: $Fe \longrightarrow Fe^{2+} + 2e^{-}$

And at the cathode hydrogen is produced and either enters the steel or forms hydrogen gas and bubbles off:

If H_2S is present: $2H^{+} + 2e^{-} \longrightarrow H° + H°$

If H_2S is not present: $H° + H° \longrightarrow H_2$

Hydrogen enters the steel first by adsorption onto the water to steel interface and then by being absorbed into the steel as hydrogen. The amount of hydrogen entry depends on the corrosion rate of the steel surface and the concentration of anions such as HS- that reduce the tendency to produce hydrogen gas (as shown in second equation above) and promotes the hydrogen (HO) to enter the steel. pH also has an influence on amount of corrosion damage. Basically, the term "sulphides" used in oil and gas operations includes all three water-soluble species, H_2S, HS- and S_2-, which coexist in a sulphide-water system. Shown in the ionization chart for hydrogen sulphide in figure 1, it can be observed that molecular H2S predominates in the acidic range, where the pH is below 7. In the range of 7 to 12, monovalent HS- predominates and above pH 12, divalent S2- predominates.

The illustration in figure 1 indicates that the most important practice is that when H_2S enters in alkaline mud, it reacts to form an alkaline sulphide, most often sodium sulphide.17 While elevated pH is a good way to keep H_2S from causing problems under certain conditions, it does not remove it from the fluid and any drop in pH can create a significant hazard. This then brought about the advent of sulphide scavengers in the history of oil and gas operations. The only safe method for the total removal of hydrogen sulphide or soluble sulphides is with a sulphide scavenger.

9.6.3.3 EFFECTS OF HYDROGEN SULPHIDE ON DRILLING FLUID

The first positive identification of soluble sulphide in the drilling fluids must be taken as a possible indication of H2S entering the fluid from some source. In the simplest case an increase in mud weight may be sufficient to control minor influxes. If levels persist chemical treatment will be required. This can take two forms; neutralization and scavenging. The two terms must not be confused. Neutralization of H2S in drilling fluid by caustic soda or lime is a common field approach. This alkalinity maintenance method has proved successful when small amounts of H_2S enter the drilling fluid, for example, from drilling up pore spaces containing the gas.

This is not a suitable approach to the control on influxes of Hydrogen Sulphide. The alkalinity neutralization of H_2S is instantly reversible by reduction in pH. Should the pH of the fluid decrease due to further influxes of H_2S, CO_2 or brine flows the sulphids may come out of solution as hydrogen sulphide gas. To ensure that such influxes are handled safely, it is essential to employ a sulphide scavenger. This term refers to any additive which is capable of reacting with one or more of the sulphide species and as a result, converting them to a more inert form. To be truly effective an irreversible and complete reaction must occur. As a mutual equilibrium exist between the three sulphide species in solution (H_2S, HS- and S_2-) irreversible and complete removal of one serves to removal all three.

To remove the problem, it is necessary to precipitate the sulphide as an insoluble inorganic sulphide. Zinc or iron compounds are usually chosen because of the relative inertness of the sulphids formed. Zinc source (Zinc oxide or carbonate) has in the past been the preferred treatment. However, continuous treatment with Zinc carbonate will produce significant undesirable carbonate concentration (Carbonate contamination) which can adversely effect on drilling fluid rehology and fluid loss. See the below table:

Contaminant	FV	PV	YP	Gels	FL	pH	Pm	Pf	Mf	Cl^-	Ca^{2+}
Carbonate	↑	_	↑	↑	↗	↓	↓	↗→	↓	_	↓
H_2S	↑	_	↑	↑	↑	↓	↓	↓	↓	_	↗

Quick reference for recognizing carbonate and H_2S contamination

As a precaution against Hydrogen Sulphide influxes into the drilling fluid system should be pre- treated with Zinc Source. Zinc oxide can be used as an alternative, provided that the possible environmental payloads associated with its use are appreciated. pH and alkalinity tests (Pf/Mf) are a good first indicator of Hydrogen Sulphide influx. However, the use of scavenger is used up. It can be seen from this that while a scavenger increase safety levels, it also makes detection of small Hydrogen Sulphide influxes difficult by conventional methods. Measurement of the concentration of scavenger in the mud system is possible but the analysis method is not safe for use on the rig.

The most effective environmentally acceptable scavengers for Hydrogen Sulphide in water base muds are Zinc sources like Zinc Carbonate or Oxide, primarily because of their solubility which leads to very fast reaction rates compared to other scavengers.

Zinc Sources clearly reacts very much faster than the other scavengers. This is due to the fact that the iron in solution ensures a fast reaction with the sulphides and the chemistry in this case produces elemental sulphur as the reaction product. Elemental sulphur is completely insoluble in water. The iron is reduced from Fe+3 to Fe+2 in the process and some of this will be regenerated to produce Fe+3 when the mud comes in to contact with the air. Any regeneration of Fe+3 will aid in reducing product consumption without affecting is scavenging properties.

Recommended treatment levels of Zinc sources are as follows:

pre- emptive treatment is recommended at 1.0 lb/bbl treatment of a mud system in the event of gas influx should be made on the basis that 2.0 lb/bb; will treat approximately 100 ppm Hydrogen Sulphide.

9.6.3.4 TESTING/ REPORTING/ TREATMENT OF HYDROGEN SULPHIDE

The active mud system must be regularly analysed and monitored for the presence of active soluble Sulphide with the Garrett Gas train. This test must be carried out at the same time as each complete mud check, i.e.: three times per day while drilling.

Alkalinity checks should be carried out intermittently in between these full checks and any significant reduction in the pH and Mf/Pf should be followed up by testing for the presence of active soluble sulphides with the Garrett Gas Train.

For routine checks lead acetate paper can be used to give a qualitative check. Should this test be positive then it must be followed by a qualitative analysis with a Dragger tube.

The following actions must be taken after the initial positive test result:

1. Inform the Drilling Supervisor
2. Start adding H2S Scavenger to the active mud system. Active soluble sulphides can be analysed and monitored because of their characteristic of readily regenerating hydrogen Sulphide gas in reaction with mild acid. By adding citric acid to whole mud sample in a Garrett gas train detection of Hydrogen Sulphide on lead acetate paper indicates the presence of active Sulphides in the mud. When this qualitative test proves positive a dragger tube should be used to give qualitative results.

However, it must be remembered that pre-treating the mud with MILGARD Fe means that no Sulphides will be delectable until all the scavenger has reacted, i.e.: it is impossible to detect small influxes of Hydrogen Sulphide in a system pre- treating with MILGARD Fe.

9.6.4 Contingency stock to be kept on hand on the rig

The following Contingency stock of treatment chemicals must be kept on hand on the rig over and above the normal mud maintenance material requirements.

- Zinc Carbonate or Oxide 120 Sxs 25 kg
- Lime 200 Sxs 25 kg

As above mentioned Hydrogen Sulphide is a poisonous and dangerous acidic gas encountered in many formations. It can quickly deaden senses and can be fatal, even at low concentration. Hydrogen Sulphide is characterized by its typical “rotten egg” smell. For safety reasons, it should be neutralized immediately

with caustic soda or lime to increase pH to >11.5 to form sulphide (S2-), then precipitated with a source of Zinc.

Neutralization with caustic soda: $2\ NaOH + H_2S\ 2Na+ + S_2- + 2H_2O$ (at pH>11.5)

Neutralization with maintaining excess lime: $2Ca(OH)_2 + H_2S\ Ca_2+ + S_2- + 2H_2O$ (at pH>11.5)

Converting Hydrogen Sulphide into just sulphide by increasing pH is not a permanent reaction. If the pH were drop into the acid region, sulphide will be converting back into the poisonous H_2S form. For this reasons H_2S should always be precipitated by adding a source of Zinc such as Zinc Oxide.

Removal by precipitation with treatments of (ZnO): $H_2S + ZnO\ ZnS + H_2O$

if H_2S gas detected in mud following action should be taken to:

- Immediately rise up the pH to at least 11.5 to 12 with caustic soda.
- Buffer the pH with lime.
- Begin treatment with Zinc oxide to remove sulphide (S_2-) from the system. (1 lb/bbl treatment of Zinc oxide removes approximately 1100 mg/lit of sulphide. No more than 2 lb/bbl of Zinc oxide should be used to pre-treat the system for H_2S) If it appears that H_2S bleeding in (flowing) form the formation, the mud density should be increased to shut off the flow of gas into the well-bore.

9.7 Cement Program

Detailed program for spot 118 pcf cement plug (Cement Class G) from 3400 to 3575 m and Cement 5" Liner will be issued in due time.

9. Completion

Completion History:

RIH Completion String as Program to 3496m While Press. Test Comp. String Several Steps W/ 6000 PSI / 15 MIN, OK. Pressure up 500psi x 5mins,2000psi x 5mins,2900psi x 15mins to set packer at 3575.12m MT, ok. Then bleed off tubing pressure to 0psi in 5mins OK.
Landing tubing hanger in tubing head spool make sure expansion joint half open. Tight and lock all tie down screw. Pressure test surface line and tubing spool valve to 3000psi x 15mins, ok.
Retrieve plug and prong from No-Go, Apply pressure from tubing head spool side outlet valves in annulus to 2000psi for 15mins to confirm packer is set, ok.
Pressure test completion string completely to 6000 psi x 15mins, ok.
R/U slick line and connect to landing joint, test all connection with 5000psi for 15mins, ok.
RIH slick line with 2.5" SB pulling tool to 3590.08m (depth of landing nipple) to pull out the WRB-2 equalizing check valve from landing nipple, POOH to surface and L/D it. R/D slick line units. Pressure test control line 8000psi for 15mins, ok, then bleed off the pressure to 0 psi.
Break off & L/D landing joint. Install TWCV in tubing hanger. Close BOP rams, test TWCV and tubing hanger to 6000psi x 15mins, ok.
N/D Bell nipple and BOP.
N/U tree adapter and X-mas tree, feeding through X-mas tree adapter with 1/4" control line.
Test control line w/ 8000 psi, 15 min, "ok". Release C/L pressure to zero (close TRSSSV).
Install x-mass tree, pressure test x- mass tree related valves.
Pull out TWCV from tubing hanger.

10.1 Equipment Specification

ITEM	EQUIPMENT DESCRIPTION	
1	**FLOW COUPLING**	
	MAX OD:	5.250"
	MIN ID:	3.640"
	THREADS:	4-1/2" 18.90 PPF TPCQ-EX BOX
	WORKING PRESSURE:	10,000 PSI
	LENGTH:	4FT
	MATERIAL:	L80
	S/N	1512Y151105-03 /04
	COMPANY:	WEIQIXIN
	TUBING RETRIEVABLE SAFETY VALVE	
	WORKING PRESSURE:	10,000 PSI
	TEMPERATURE:	325DEG F
	MAX OD:	7.500"

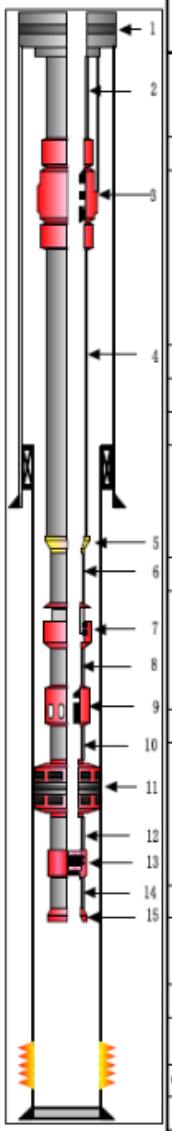
2	MIN ID:	3.437"
	PROFILE:	AF
	SELF-EQUALIZING FEATURE:	YES
	MATERIAL:	SUPPER 13CR
	THREADS:	4-1/2" 18.90 PPF TPCQ-EX BOX x PIN
	S/N	1512 15080-04
	COMPANY:	WEIQIXIN
3	**SPLINED EXPANSION JOINT**	
	STROKE LENGTH:	10FT
	SHEAR PIN SET:	24000 LBS (WELL SITE ADJUSTABLE)
	WORKING PRESSURE:	10,000 PSI
	TEMPERATURE:	325DEG F
	TOOL OD:	4.906"
	TOOL ID:	2.922"
	THREADS:	3-1/2"10.20 PPF TPCQ-EX BOX x PIN
	MATERIAL:	SUPPER 13CR
	TRANSFER TORQUE FEATURE:	YES
	S/N	1512 15121-04
	COMPANY:	WEIQIXIN
4	**SLIDING SIDE DOOR**	
	PROFILE:	X
	MAX.OD:	4.551"
	MIN ID:	2.750"
	WORKING PRESSURE:	10,000 PSI
	TEMPERATURE:	325DEG F
	THREADS:	3-1/2" 10.20 PPF TPCQ-EX BOX x PIN

ITEM	EQUIPMENT DESCRIPTION	
	MATERIAL:	SUPPER 13CR
	OPEN METHOD:	DOWNWARD OPEN
	S/N	1510 15084-03
	COMPANY:	WEIQIXIN
5	**LANDING NIPPLE**	
	PROFILE TYPE:	XN
	WORKING PRESSURE:	10,000 PSI
	MAX OD.:	4.250"
	SEAL BORE ID:	2.562"
	MIN ID.:	2.442"
	THREADS:	3-1/2" 10.20 PPF TPCQ-EX BOX x PIN
	MATERIAL:	SUPPER 13CR
	S/N	1605Y160510-05

	COMPANY:	WEIQIXIN
6	**RE-ENTRY GUIDE SHOE**	
	MAX OD:	4.250"
	MIN ID:	2.900"
	TOP CONNECTION	3-1/2" 10.20 PPF TPCQ-EX BOX
	MATERIAL:	SUPPER 13CR
	S/N	1601Y151112-04
	COMPANY:	WEIQIXIN
7	**FITTING PIPE**	
	TOP CONNECTION	1/4” NPT PIN X 1/4” JAM NUT CONNECTOR
	MATERIAL:	INCOLOY825
	S/N	SW025- JN002
	COMPANY:	WEIQIXIN
8	**HYDRAULIC SET PERMANENT PACKER**	
	TYPE:	PERMANENT & SINGLE TRIP
	INTIAL SET PRESSURE	1800PSI
	MAX SETTING PRESSURE:	2900PSI
	WORKING PRESSURE:	10,000PSI
	MAX OD:	5.750"
	MIN ID:	2.922"
	TEMPERATURE:	325DEG F
	TOP CONNECTION:	LEFT HAND SUIT WITH THE ANCHOR SEAL
	BOTTOM CONNECTION:	3-1/2" 10.20 PPF TPCQ-EX PIN
	MATERIAL:	S13CR-110KSI
	ELEMENT MATERIAL:	AFLAS
	S/N	1511 15078-07
	COMPANY:	WEIQIXIN
9	**ANCHOR SEAL ASSEMBLY**	
	WORKING PRESSURE:	10000 PSI
ITEM	**EQUIPMENT DESCRIPTION**	
	TEMPERATURE:	325DEG F
	MIN ID.:	2.922"
	TOP CONNECTION:	3-1/2" 10.20 PPF TPCQ-EX BOX
	BOTTOM CONNECTION:	MULE SHOE GUIDE
	MATERIAL:	S13CR-110KSI
	TYPE:	RATCH LATCH
	S/N	1510 15079-04
	COMPANY:	WEIQIXIN

10.2 First Run Completion String Which was Failed (Weqixin Co.)

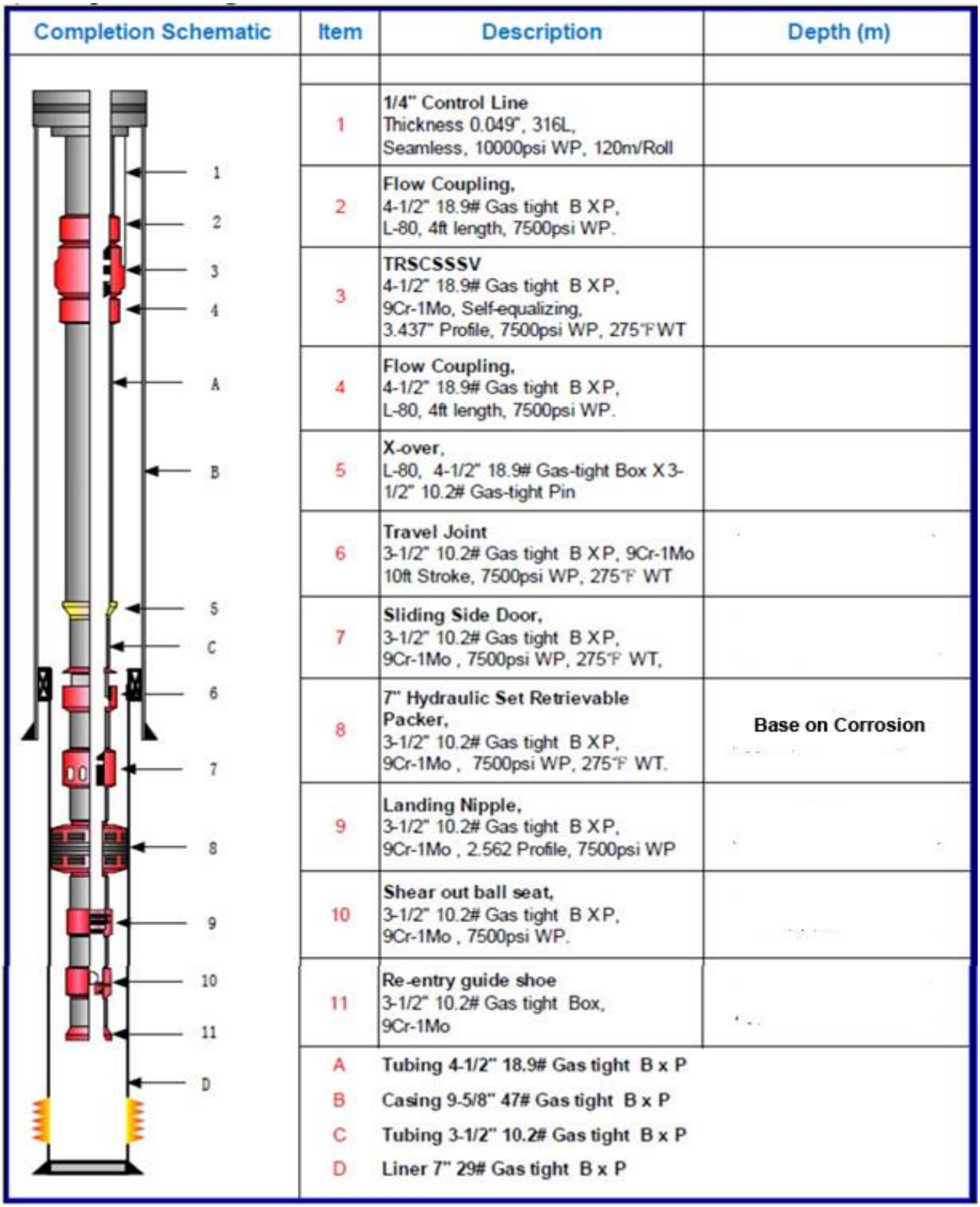
Completion String Schematic of AZNS-075



Item	DESCRIPTION	ID IN	OD IN	LENGTH M	TOP DEPTH M	BOTTOM DEPTH M
1	TUBING HANGER,11" BOWL X 4-1/2"-4 ACME-2G-LH THREAD TOP AND 4-1/2" TPCQ-EX BOX BOTTOM WITH 5-1/2" EXTENDED NECK.	3.640	11.000	0.370	10.690	11.060
	Double-Pin Joint, 4-1/2"18.90 PPF TPCQ-EX P × P	3.640	4.500	1.500	11.060	12.560
2	Tubing, 4-1/2" 18.90 PPF TPCQ-EX BOX x PIN (9 joints)	3.640	4.500	84.462	12.560	97.022
3	Pup Joint, 4-1/2"18.90 PPF TPCQ-EX B × P	3.640	4.500	1.730	97.022	98.752
	FIOW COUPLING,4-1/2" 18.90 PPF TPCQ-EX BOX	3.640	5.250	1.120	98.752	99.872
	TRSSV-10 TUBING RETRIEVABLE SURFACE CONTROLLED SAFETY VALVE 4-1/2" 18.90 PPF TPCQ-EX BOX x PIN	3.437	7.500	1.740	99.872	101.612
	FIOW COUPLING,4-1/2" 18.90 PPF TPCQ-EX BOX	3.640	5.250	1.120	101.612	102.732
	Pup Joint, 4-1/2"18.90 PPF TPCQ-EX B × P	3.640	4.500	1.730	102.732	104.462
4	Tubing 4-1/2" 18.9PPF TPCQ-EX BOX x PIN(227 joints)	3.640	4.500	2154.810	104.462	2259.272
5	CROSS-OVER, 4-1/2" 18.90 PPF TPCQ-EX BOX x 3-1/2"10.2ppf TPCQ-EX PIN	2.922	5.250	0.940	2259.272	2260.212
6	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(134 joints)	2.922	3.500	1280.285	2260.212	3540.497
7	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.745	3540.497	3542.242
	3-1/2" X 10FT SPLINED EXPANSION JOINT 10.20 PPF TPCQ-EX BOX x PIN	2.922	5.000	5.770	3542.242	3548.012
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.745	3548.012	3549.757
8	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	3.500	9.570	3549.757	3559.327
9	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.740	3559.327	3561.067
	SSD 2.75" NON-ELASTOMERIC SLIDING SIDE DOOR 3-1/2" 10.20 PPF TPCQ-EX BOX x PIN	2.922	5.000	1.195	3561.067	3562.262
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.745	3562.262	3564.007
10	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	3.500	9.583	3564.007	3573.590
11	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.745	3573.590	3575.335
	7" 29-35 PPF CASING HYDRAULIC SET PERMANENT PRODUCTION PACKER WITH SIZE 4.00" ANCHOR SEAL ASSEMBLY 3-1/2" 10.20 PPF TPCQ-EX BOX x PIN	2.922	5.745	1.805	3575.335	3577.140
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.745	3577.140	3578.885
12	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	3.500	9.583	3578.885	3588.468
13	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.745	3588.468	3590.213
	2.562"LANDING NIPPLE 3-1/2" 10.20 PPF TPCQ-EX BOX x PIN	2.635	4.250	0.385	3590.213	3590.598
14	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	3.500	9.585	3590.598	3600.183
15	RE-ENTRY GUIDE SHOE 10.20 PPF TPCQ-EX BOX UP X MULE SHOE GUIDE DOWN	2.900	4.250	0.210	3600.183	3600.393
Other						
A	1/4" X 0.065" X 120 METERS CONTROL LINE					

Figure 9 2nd Completion Schematic

10.3 Proposed Completion string



Completion Schematic	Item	Description	Depth (m)
	1	1/4" Control Line Thickness 0.049", 316L, Seamless, 10000psi WP, 120m/Roll	
	2	Flow Coupling, 4-1/2" 18.9# Gas tight B X P, L-80, 4ft length, 7500psi WP.	
	3	TRSCSSSV 4-1/2" 18.9# Gas tight B X P, 9Cr-1Mo, Self-equalizing, 3.437" Profile, 7500psi WP, 275°F WT	
	4	Flow Coupling, 4-1/2" 18.9# Gas tight B X P, L-80, 4ft length, 7500psi WP.	
	5	X-over, L-80, 4-1/2" 18.9# Gas-tight Box X 3-1/2" 10.2# Gas-tight Pin	
	6	Travel Joint 3-1/2" 10.2# Gas tight B X P, 9Cr-1Mo 10ft Stroke, 7500psi WP, 275°F WT	
	7	Sliding Side Door, 3-1/2" 10.2# Gas tight B X P, 9Cr-1Mo , 7500psi WP, 275°F WT,	
	8	7" Hydraulic Set Retrievable Packer, 3-1/2" 10.2# Gas tight B X P, 9Cr-1Mo , 7500psi WP, 275°F WT.	Base on Corrosion
	9	Landing Nipple, 3-1/2" 10.2# Gas tight B X P, 9Cr-1Mo , 2.562 Profile, 7500psi WP	
	10	Shear out ball seat, 3-1/2" 10.2# Gas tight B X P, 9Cr-1Mo , 7500psi WP.	
	11	Re-entry guide shoe 3-1/2" 10.2# Gas tight Box, 9Cr-1Mo	
	A	Tubing 4-1/2" 18.9# Gas tight B x P	
	B	Casing 9-5/8" 47# Gas tight B x P	
	C	Tubing 3-1/2" 10.2# Gas tight B x P	
	D	Liner 7" 29# Gas tight B x P	

Figure 11: Proposed Completion Schematic

10. BOP, Wellhead & X-Mass Tree

11.1 Existing Wellhead & X-Mass Tree

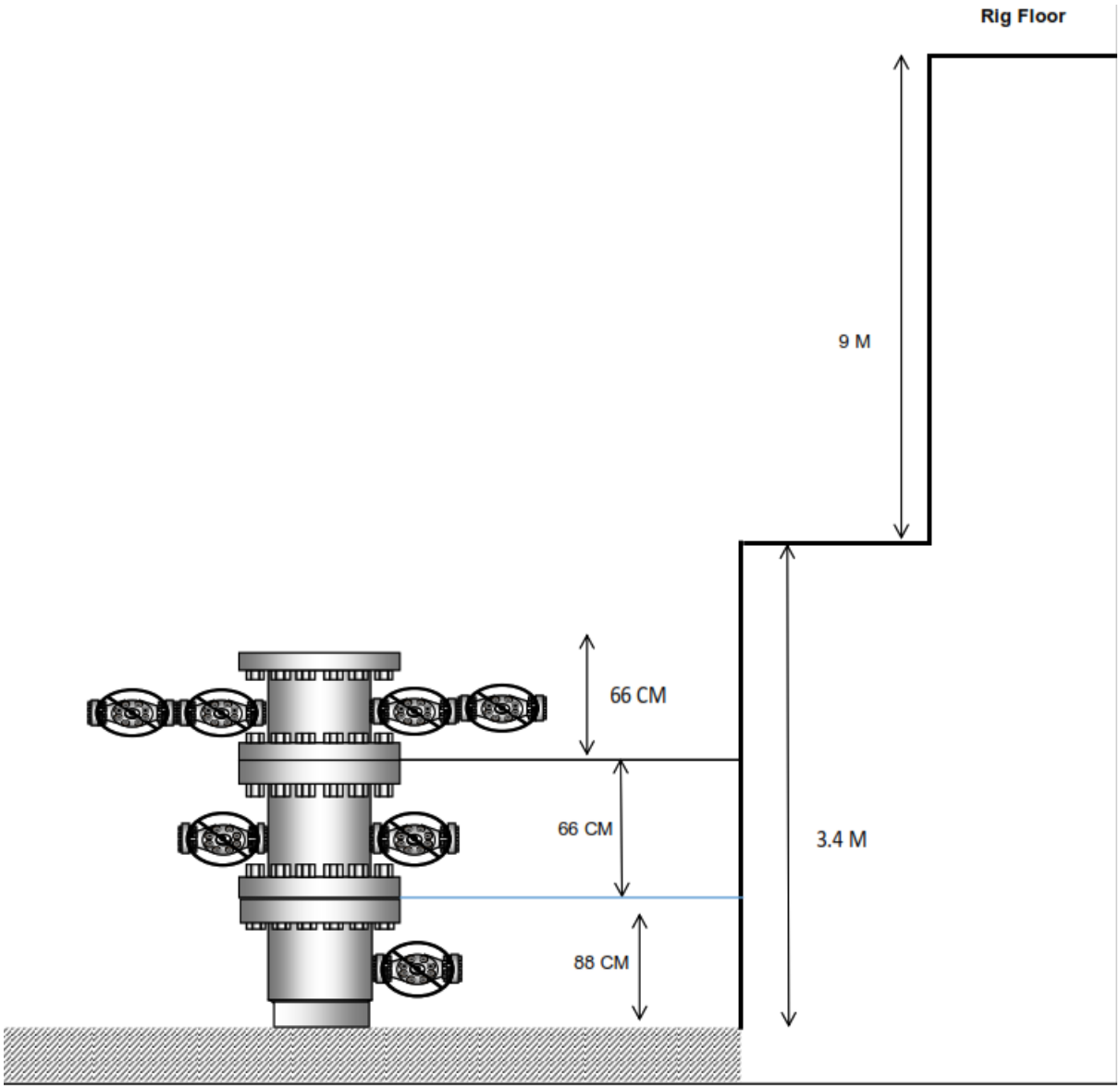


Figure 12: Existing Wellhead

11.2 Casing head housing (20 3/4"-3K)

Item	Description (Equipment and ancillary tools)	S/N	M/C	T/C	PSL	Qty	Manufacture
1	20-3/4" 3K Casing head housing	CH1416-S140007	DD-NL	U	2	1 set	MSP
2	Gate valve assembly 2-1/16" 5000psi	N150084	DD-NL	U	2	1 set	MSP
3	Number of Installed VR Plugs	0					MSP
4	Number of Installed Companion Flanges	2 sets					MSP

11.3 Casing Head Spool (20-3/4" 3K * 13-5/8", 5K)

Item	Description (Equipment and ancillary tools)	S/N	M/C	T/C	PSL	Qty	Manufactur
1	Casing head spool 20-3/4" 3000psi x 13-5/8" 5000psi	CS1128 N130025	DD-NL	U	3	1 set	MSP
2	Casing Hanger					1 set	MSP
3	Gate valve assembly 2-1/16" 5000psi	N130689	DD-NL	U	3	1 set	MSP
4	Gate valve assembly 2-1/16" 5000psi	N130673	DD-NL	U	3	1 set	MSP

Item	Description (Equipment and ancillary tools)	S/N	M/C	T/C	PSL	Qty	Manufacture
5	Number of Installed VR Plugs	0					MSP
6	Number of Installed Companion Flanges	2 sets					MSP

11.4 Tubing head spool (13-5/8", 5K x 11"10K)

Item	Description (Equipment and ancillary tools)	S/N	M/C	T/C	PSL	Qty	Manufacture
1	Tubing head spool 13-5/8", 5K x 11"10K	N150005	EE-0.5	U	3	1 set	MSP
2	Tubing Hanger[1]	2909423B N150002	FF-0.5	U	3G	1 set	MSP
3	Gate valve assembly 2-1/16" 10000PSI	N140112	EE-0.5	U	3	1 set	MSP
4	Gate valve assembly 2-1/16" 10000PSI	N140133	EE-0.5	U	3	1 set	MSP
5	Gate valve assembly 2-1/16" 10000PSI	N140111	EE-0.5	U	3	1 set	MSP
6	Gate valve assembly 2-1/16" 10000PSI	N140098	EE-0.5	U	3	1 set	MSP

Item	Description (Equipment and ancillary tools)	S/N	M/C	T/C	PSL	Qty	Manufacture

7	Number of Installed VR Plugs	0	MSP
8	Number of Installed Companion Flanges	2 sets	MSP

11.5 Section D

Item	Description (Equipment and ancillary tools)	S/N	M/C	T/C	PSL	Qty	Manufacture
1	BTM Master Valve with Tubing Head Adaptor^2 11"x10k-4-1/16x10K	N150007	FF-0.5	U	3G	1 set	MSP
2	Top Master Valve 4-1/16", 10000 Psi	008929J-N150009	FF-0.5	U	3G	1 set	MSP
3	Safety Valve 4-1/16" 10000psi	0269161B N150009	FF-0.5	U	3G	1 set	MSP
4	SSV Actuator, 6000 psi	026916B N150009	AA	U	3G	1 set	MSP
5	Gate valve assembly 4-1/16" 10000psi (Swab Valve)	008929H N150009	FF-0.5	U	3G	1 set	MSP
6	Gate valve assembly 4-1/16" 10000psi (Kill Line Valve)	002929A N150055	FF-0.5	U	3G	1 set	MSP

Item	Description (Equipment and ancillary tools)	S/N	M/C	T/C	PSL	Qty	Manufacture
7	Gate valve assembly 4-1/16" 10000psi (Production Line)	002929A	FF-0.5	U	3G	1 set	MSP

	Valve)	N150056					
8	Manual Adjustable Choke Valve 4-1/16" 10000psi	0139179 N150009	FF-0.5	U	3G	1 set	MSP
9	Solid Block 4 1/16” 10000 psi	N150009	FF-0.5	U	3G	1 set	MSP
10	Number of Installed VR Plugs	0					MSP
11	Number of Installed Companion Flanges	2 sets					MSP

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Sarvak	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Kazhdumi/Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K

Item	Specification
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Sarvak & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Wellhead Specification 20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Gadvan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange and 2" bull plug threaded for ½" NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange and 2" bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130**,** Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange and 2" bull plug on one side and, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange and 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm** Material of body: Forged low alloy steel 410SS G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

11.6 BOP

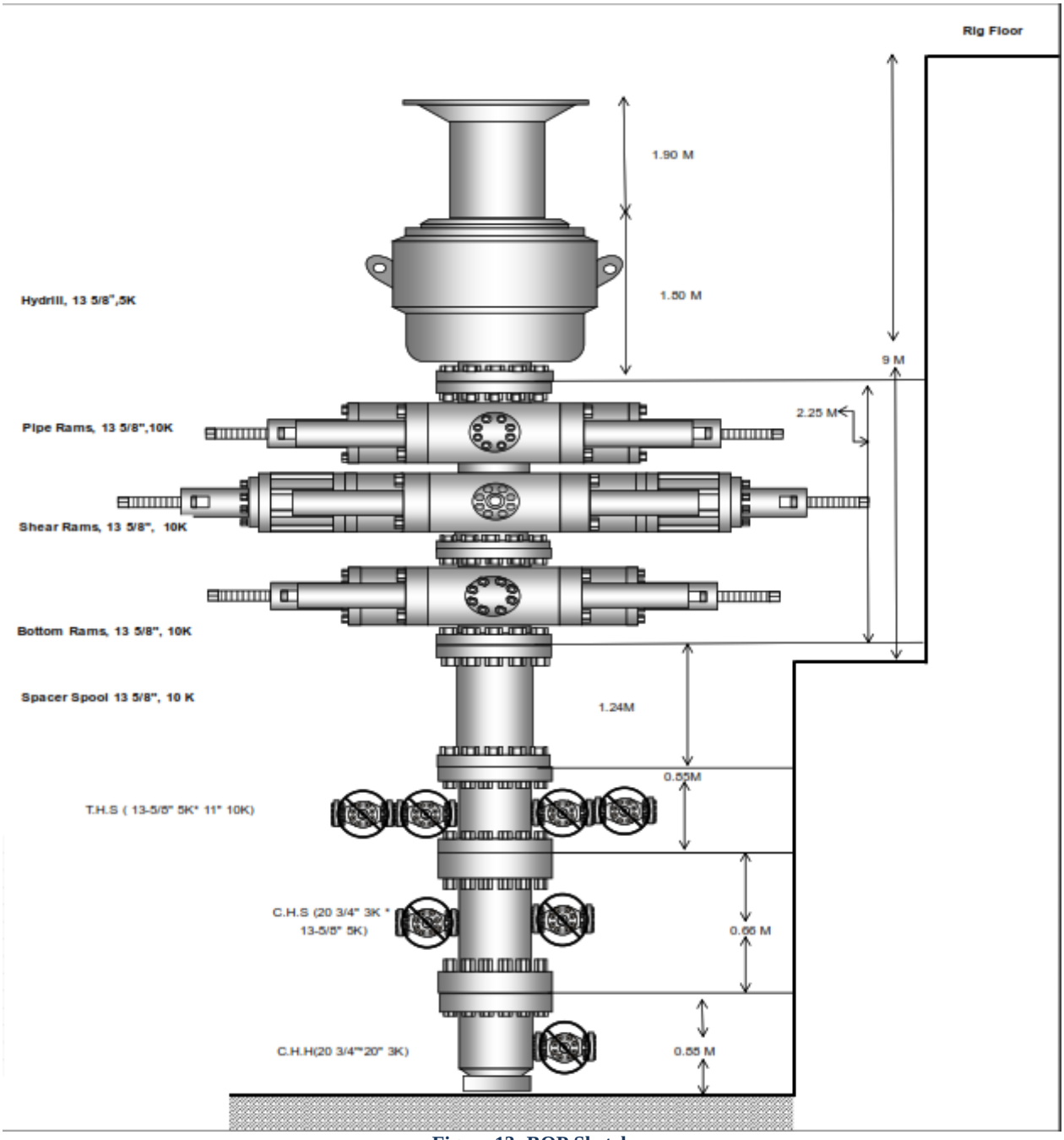


Figure 13: BOP Sketch

11.

11. Waste Management

There is drilling objective and drilling cutting in AZNS-075 workover well after Side Tracking; therefore waste management program will be issued in due time.

In case of requirement to dispose wastewater in workover well, waste management program will be issued at due time as PEDEC policy.

12. Health, Safety and Environments

13.1 WELL CONTROL

All the drilling operation shall obey the requirement of Well Control Manual of MSA and contractor's well control manual; under no circumstances should the deviation from well control policy be permitted without prior approval through MSA. If the contractor's well control manual or anywhere has conflict with Well Control Manual of MSA, take Well Control Manual of MSA as the standard.

13.1.1 WORK Requirement of Well Control Equipment

- With shut-in casing pressure lower than 1,000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2,000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

13.1.2 Management of Well Control Equipment

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

13.2 Recording Reference

All BOP tests, drills, function tests, any malfunctions, repair or maintenance to the mud System and well control equipment shall be recorded in the IADC daily reports and shall be signed by both the Drilling Contractor's Tool Pusher and Drilling Supervisor on the well site. They shall also be recorded in the Daily Drilling Report.

13.3 Flow Check Procedure

To avoid a kick situation, it is recommended that a flow check be conducted at specific times during the drilling operation.
While drilling (any warning sign).
Prior to tripping the drill string.
While tripping (any warning sign).
While out of hole (continuously).
Trip out:
After pulling 5% of the drill pipe
At the midpoint depth of the wellbore
Prior to pulling the first stand of drill collars
After all the drill string is out of hole.
Trip in:
Upon reaching surface casing shoe
At midpoint of the wellbore

13.4 Primary Control Reference

It is the responsibility of the Drilling Contractor to keep the hole full at all times by using all the available equipment. The Drilling Supervisor shall monitor that the correct preventive practices are being implemented and/or conducted at all times.
Extreme care shall be taken to monitor mud volume, drilling breaks and gas cut mud.
The MACP (Maximum allowable casing pressure) shall be confirmed by the Drilling Supervisor, either after setting each new casing string or, whenever the density of the drilling mud changes.
The MACP shall be clearly written on a kick control sheet that will be posted near the choke control panel.
The predetermined distance at which the Tool Joint is to be pulled above the rotary should be posted on rig floor near the BOP control panel.
The Driller is responsible for carefully measuring and recording the slow pump pressure.
Awareness of these values is an important element in killing operations, in order to avoid formation breakdown.

Take a measure and record the slow pump pressure at three different flow rates with all pumps at the following times:
Once per tour.
Whenever changes occur in the mud density/rheology.
Whenever changes occur in the dimension and characteristics of the string, BHA, jet size, jet plugged or jet lost, etc.

13.5 Secondary Control Reference

Any time a drilling break is noticed, drilling shall stop and a static monitoring of the well shall be carried out. Space-out for all Pipe Rams and Annular Preventer must be known to the Driller at all times to keep tool joints clear of annular preventer and rams.

13.6 Well control during drilling and completion operation

The drilling design shall be strictly implemented by the drilling contractor and when it is discovered there is incompliance of downhole situation with design and the design will be amended, the procedure for amendment of design shall be followed. If emergency is encountered, and safe operation can be affected, after the discussion between drilling rig manager and the drilling supervisor, practical technical measures can be taken first and then report to the top management. The Company shall approve and notice in time as per the drilling program amendment procedure.
The curve of well equivalent mud weight as per predicted formation pressure, curve of mud weight in drilling program and curve of actual mud weight must be drawn during drilling operation. For exploratory well and key well, the curve of dc index while drilling formation pressure equivalent mud weight for sandstone, mudstone and shale shall be drawn by the comprehensive mud logging crew.
When entering oil and gas formation or drilling master reservoir, high diameter nozzle shall be selected for the drilling bit to treat well killing and mud loss.
SCR (Slow Circulation Rate) shall be conducted at the start of drilling activity, pump strokes, pumping rate and circulating pressure should be recorded. While drilling in pay zone, SCR is required to be conducted with all pumps separately as follows, pump strokes, pumping rate and circulating pressure shall be recorded in Daily Drilling Report and Kill Sheet.
Once for every 12 hours in general;
After bit change or BHA change; 3) Mud weight change exceeds $^{3.75}$pcf.
Oil and gas invasion and overflow shall be checked with wiper trip and shall be conducted under the following situation:
Each time before POOH after drilling oil and gas formations.
Severe invasion of oil and gas occurs during drilling, but overflow does not occur before POOH.
POOH after killing well.
POOH after lowering mud weight in wellbore.
POOH after curing mud loss during drilling oil and gas formation.
POOH after drilling oil and gas formation and a prolonged period of time is needed for other operations (E logging, running casing, running tubing and DST etc.).
Regarding the shut-in methods selection
Soft shut-in method is recommended for gas well;
Hard shut-in method is recommended for oil well;
Hard shut-in method is recommended for the well of which the well has strong formation strength in EMW as per FIT test;

Soft shut-in method is recommended for the well of which the well has weak formation strength in EMW as per FIT test.

Main technical measures for prevention of kicking and blowout during tripping.

Before POOH, circulation duration of mud in wellbore shall not be less than one and a half cycles. The observation time after wiper trip shall be no less than one and a half cycles. The density differential of inlet and outlet shall not exceed 1.248 pcf.

Flow check shall be conducted regularly via trip tank as follows

Prior to POOH (keep string rotation at 10 rpm for preventing stuck pipe)

POOH to last casing shoe

Before POOH BHA through BOP stack

Note: Tripping should be shut-off immediately and flow check shall be performed later on when any suspected well kick is observed.

During POOH, fill up hole every 3-5 stands of DP and every stand of DC. If the nozzle is plugged, during POOH, mud shall be filled full for each joint or continuous filling shall be conducted and trip sheet shall be recorded carefully.

When the bit is in the oil and gas formation and 300m above top of oil and gas formation, POOH speed shall be controlled to maintain the excellent well building property and rheological property of mud to avoid severe swabbing in the wellbore.

During RIH, the running speed shall be controlled to avoid the loss of circulation due to down hole pressure surge.

During tripping, check and repair of the equipment shall be arranged at the time when running to the casing shoe. if equipment have to be checked and repaired during POOH, due blowout prevention measures shall be taken during and after the check and repair, drill string shall be run to bottom and circulate for one and a half cycles. After everything is normal, drilling shall be resumed. Equipment check and repair shall not be conducted for empty well condition.

Treatment measures for invasion of gas during normal drilling:

When invasion of gas is discovered, it shall be discharged with circulation in time. The gas invaded mud shall not be pumped in the hole before eliminating gas.

Increasing mud weight shall be conducted when stop drilling and after invaded gas eliminated from mud. Mud weight incremental of each circulation cycle shall be controlled within 3.121 pcf

"In-post" policy

"In-post" will be initiated after the surface casing is cemented.

The "in-post" personnel must be trained and verified qualified by the technical staff of drilling rig.

During drilling, specific personnel shall be arranged to be "in-post", the mud logging personnel shall monitor the level of the circulating tank etc. as per the requirements. During tripping, other auxiliary operation or drilling suspension, specific personnel for "in-post" shall be arranged by the drilling and mud logging personnel in the meantime.

"In-post" requirements:

For drilling operation, variation of the outlet flow, drilling time, lithology, bubble, smell, showings, pit volume, density and mud viscosity, gas measured values and content of chloride etc. shall be carefully monitored. The level of the circulating tank shall be observed and recorded each 15 minutes and if special situation occurs, the measurement and record shall be intensified. If anomaly occurs, it shall be reported to the driller in time.

For tripping operation, trip sheet shall be filled accordingly. If anomaly occurs, it shall be reported to the driller.

For other operations, it shall be observed whether there is overflow of mud at the wellhead or mud level from wellhead. If anomaly occurs, it shall be reported to the driller.

Treatment of loss of circulation

Before cementing non-target zone or pay zone, if loss of circulation occurs during drilling, POOH shall be carried to run open ended drill pipe for cement plug job if condition allows. During RIH, mud shall be filled from annulus to balance formation pressure to avoid blowout.

Treatment of loss of circulation during drilling after production casing cemented

Mud level not at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and observe with moving the drill string and prepare well control related work. Each 10 minutes, mud with density no lower than the drilling weight shall be reversely filled up from the annulus. Each time, the filling up shall be 1~2m^3, or continuous reserve injecting mud with small speed to maintain that the column pressure of mud in wellbore is higher than the formation pressure. POOH to casing shoe or safety interval for observation. Additional treatment shall be made as per the situation of loss of circulation.

Mud level at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and move the drill string to observe with well control well prepared. When the well is static, pull the bit upward to the casing shoe or safe interval for observation. If no anomaly is observed, circulate with slow pump rate. The measures such as plugging back job, POOH or condition mud etc. shall be carried out as per the situation of loss of circulation. If loss of circulation occurs to pay zone, mud weight shall be acquired by static pressure. After pulling out to casing shoe and observing for one tripping cycle, when no more loss of circulation occurs after running to bottom, drilling operation can be resumed.

Well control for E logging, cementing and DST

Before E logging, down hole condition shall be stable and static. If it takes long time for electrical logging, one more condition trip shall be conducted between logging. Shooting nipple is required for logging. Logging service shall be furnished with tools for cutting the cables.

Continuously fill the well during performing wireline logging, and keep closely monitoring for the well via trip tank.

Before running casing, ensure the well is in static condition without any flowing. Pipe ram shall be changed to casing ram with proper size. Casing circulating head or crossover sub between casing and in used DP shall be kept at rig floor at all times during casing running.

The hydrostatic pressure shall be kept balance with formation pressure during whole cementing job to avoid blowout due to loss of circulation, loss of balance for pressure and cement weight loss during WOC.

13.7 HEALTH

This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.

A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.

An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centres manned by qualified doctor 24-hours per day in Ahwaz city.
Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.

13.8 SAFETY

Due to high concentration of H2S in Sarvak, Kazhdumi and Gadvan reservoirs, Make detail H2S operation procedures, in particular, and carry it out strictly.
Lifesaving equipment like SCBA, Escape Mask, H2S detectors (Multi portable and personal detector) and implement the H2S Services system during drilling activity should be provided.
All lifting equipment and cranes must be certified by approved 3rd party (approved by Institute of Standards & Industrial Research of Iran)
All electrical equipment should be designed for such locations, and listed by nationally recognized testing laboratory. All wiring components and electrical equipment should be maintained in accordance with the manufacturers' recommendation.
PSSR (Pre-Start-up Safety Review) before spud-in and penetrating into oil and gas formations are mandatory by contractor under the supervision of Company and approved by Company.

13.9 ENVIRONMENT

Requirement on location

The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.
Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.
Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.
Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.
Contractor shall perform on-site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation Contractor in accordance with the following waste hazard classification.
Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)
Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass.
Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.
The Contractor/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.

It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to.
Local community are recommended.

Hazardous Wastes

Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site

Mud purifying circulation system.
Cooling water spraying and circulating system for drilling pump.
Special recovery tank for deposed oil.
Reserve house for valuable mud chemicals.
Automatic mud weighting supply system.
BOP stack is equipped according to drilling design.
Define a separation area for gathering all waste according to waste management procedure

Requirement of environment protection on location

Sewage pit Area of the sewage pit: 1500 m2.

Prior to spud, mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.
Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.
Install oil supply lines and check whether there is any leakage.

Management of environment protection on location

In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.
Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.
Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.
Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.
After the well is completed, reinforce the sewage pit solidify the disposed mud and have environment procedure fulfilled.
Install fuel supply lines and keep them in good condition.
All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.
The environmental protection personnel from contractor and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.

Transition Area for Hazardous Waste shall also be provided by each Contractor in his allocated area, Company shall approve each area before use as well. The transition area of hazardous waste shall be:
Paved, fenced, marked and illuminated
Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g. Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).
In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).
Configured to accommodate a safe physical separation of incompatible waste materials (e.g. acids, and alkalis, organic materials and strong oxidizing agents).
Capable enough to provide storage for the entire range of hazardous wastes generated during the work.
The area shall be provided with Spill Containment and means of preventing run-off to and from the area.
All waste containers to be stored shall be appropriately labelled.
Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from Company and/or PEDEC.

13. H_2S Monitoring & Prevention Measure

14.1 Monitoring H2S

Somewhere on the well site is apt to accumulate H2S, such as mud pit, shale shaker, and drilling platform, should be installed one set of. Make sure H2S monitor and alarm system can be used. Alarm threshold setting for H2S warning indicator: the first warning level is 10mg/m3, unable warning sound, just show present value of H2S content to operators. The second warning level is 20mg/m3, when the H2S content in air exceed 20mg/m3, H2S warning indicator can give an alarm automatically, and warning sound should be heard by everyone work in well site. The audible alarm should be installed on Kelly board.

Equip enough gas masks on drilling platform, shaker screen (total amount should large than 24), and equip 12 sets of 30min positive pressure Air Aspirators 2 sets of portable H2S monitor and appropriative H2S warning indicator. Arrange a person specially assigned for taking charge all the protective apparatus. This man must do periodical check to keep these equipment in a good condition.

H2S monitor should be done weekly check and compulsory examination.

After drilling into the pay zone, strengthen the detection of H2S content in mud.

14.2 Measures for H2S prevention

Windsocks shall be set on the drill floor and in the monsoon direction or/and prevailing wind direction of Azadegan oil field, off well site. In emergency, employees must follow the evacuation procedure.

Explosion-proof blowers shall be installed around the rig, shale shaker, and drilling fluid tanks to blow away toxic and flammable gas at the well site.

The rig crew in the H2S area shall be furnished with H2S monitors and adequate protective apparatus. Everybody shall be able to use, maintain, and check the apparatus.

The technicians of rig crew are responsible for H2S proof training and the tool pusher is in charge of the supervision and examination.

The pH value of drilling fluid in the H2S area shall be kept over 9.5. Pay attention to measurement of H2S concentration in drilling fluid. The H2S concentration shall be kept at less than 50mg/cm3 by using sulphur removal agents and degassers.

Personnel safety and protection shall be carried out according to SY/T5087 during operations in polluted areas where H2S content exceeds the critical concentration.

When the rig crew makes decision to bleed and fire because well control cannot be performed due to well site conditions, operations shall be performed according to procedures in SY/T5087.

Once blowout occurs, the incident shall be reported to the higher level department as soon as possible and firefighting trucks, ambulances, medical care personnel and technicians shall be on duty at the well site.

After the blowout is controlled, H2S concentration should be measured in the places where H2S is possible to gather.

The personnel are not allowed to access until the H2S concentration is below the critical concentration.

14. Handover Data

The handover document should contain the following well construction information:

- Pre-spud Inspection Report
- Daily Drilling Report
- Daily Mud Report
- Daily Mud Logging Report
- Completion Tally
- Completion running Report
- TMS data (Completion)
- Dry Test report
- Test Charts, X-Mass tree Installation & test Report.

15. Requirements

Item	Description	QTY.	Unit	Supplier
1	TBG hanger Running tools	1	Ea.	PEDEC-PPL
2	BPV with inserting & retrieving tools	1	Ea.	PEDEC-PPL
3	3SV Dummy Valve inserting & retrieving tools	1	Ea.	PEDEC-PPL
4	3SV Lock-Open Tools with inserting & retrieving tools	1	Set	PEDEC-PPL
5	Separation Sleeve for 3SV with inserting & retrieving tools	1	Ea.	PEDEC-PPL
6	XOS 4-1/2" TPEX-CQ pin * IF Box	1	Ea.	PEDEC-PPL
7	XOS 3-1/2" TPEX-CQ pin * IF Box	1	Ea.	PEDEC-PPL
8	ONE Way & TWCV with inserting & retrieving tools	1+1	Ea.	PEDEC-PPL
9	Plug for No-Go Landing Nipple 2.562" Complete with Running & Pulling Tools.	1	Ea.	PEDEC-PPL
10	Prong for No-Go Landing Nipple 2.562" Complete with Running & Pulling Tools.	1	Ea.	PEDEC-PPL
11	Protection Sleeve for Tubing Retrievable Safety Valve Complete with Running & Pulling Tools Min ID 3.437", Profile AF	1	Ea.	PEDEC-PPL
12	Lubricator for BPV	1	Ea.	PEDEC-PPL

13	Shifting Tools for SSD, ID: 2.75"	1	Ea.	PEDEC-PPL
14	Plug for SSD 2.75" complete with running & pulling Tools	1	Ea.	PEDEC-PPL
15	Prong for SSD 2.75" with inserting & retrieving tools	1	Ea.	PEDEC-PPL
16	Protector for 3-1/2" Tubing	150	Ea.	PEDEC-PPL
17	Protector for 4-1/2" Tubing	280	Ea.	PEDEC-PPL
18	8-1/2" Tri Cone Bit IADC Code:537	1	Ea.	MSA
19	8-1/2" PDC Bit	1	Ea.	MSA
20	6-1/8" Tri Cone Bit IADC Code:537	2	Ea.	MSA
21	6-1/8" PDC Bit	1	Ea.	MSA
22	Single Trip Whip-stock to Set in 7" Liner 29 ppf, L-80	1	Ea.	MSA
23	Single Trip Whip-stock to Set in 9-5/8" Casing 47 ppf, L-80	1	Ea.	MSA
24	6" Mill Metal Muncher	4	Ea.	MSA
25	4-1/8" Mill/Bit	4	Ea.	MSA
26	Taper Tap Suitable for MODEL PREMIER PRODUCTION PACKER (Catch Size 2" to 4") SIZE: 7"OD:5.745” MIN ID:2.922”	2	Ea.	MSA

27	8-3/8" Metal Muncher Mill	2	Ea.	MSA
28	Gadvan Completion String 7.5K	1	set	PEDEC-PPA
29	Liner Top Packer Assembly, 5" 18-20#, L80, Premium connection, C/W Seal Nipple & Packer	1	Ea.	MSA
30	Rotating liner hanger, hydraulic set, tandem cone, with integral/split, weight set top packer, 5" 18# x 7" 29 32#, L080,110S,PREMIUM connection, 10 ft PBR, load capacity 2,000KN, packer seal capacity 10,000 psi, Temperature 150 ºC	1	Set.	MSA
31	Float shoe, reamer type, 5 " 18# ,110S,PREMIUM connection, PDC drillable	2	Ea.	MSA
32	Float collar, 5" 18# , 110S, PREMIUM connection, PDC drillable	2	Ea.	MSA
33	Landing collar c/w setting ball, 5" 18#, 110S,PREMIUM connection, PDC drillable	2	Ea.	MSA
34	Drift Gauge, API, for 4 1/2" 18.9# tubing	2	Ea.	MSA
35	Drift Gauge, API, for 5" 18# casing	2	Ea.	MSA
36	Drift Gauge, API, for 3 1/2" 10.2# tubing	2	Ea.	MSA
37	Thread Seal Compound, 10 kg/pail	10	Ea.	MSA
38	Thread Lock Compound, API modified, each box contains 12 kits, 1 lb/kit	4	Ea.	MSA
39	Stop Collar, 5", c/w nails and pins	150	Ea.	MSA

40	Centralizer, Rigid, 5" x 8 1/2"	15	Ea.	MSA
41	Centralizer, 5 " x 8 1/2", Non weld, Bow spring type, c/w nails	60	Ea.	MSA
42	Tubing pup joint, 3 1/2", L080,premium Connection 10.2 ppf,13CR01 MO, seamless,3 m long , PSL3	15	Ea.	MSA
43	Tubing pup joint, 4 1/2", L080,premium Connection 18.9 ppf,13CR01 MO, seamless,3 m long , PSL3	5	Ea.	MSA
44	Crossover, 4 1/2" 18.9 PPF Box × 3 1/2" 10.2 PPF Pin, Gas Tight Conn, L080 type1 1.2m long, PSL3.	5	Ea.	MSA
45	Tubing pup joint, 3 1/2", L080 type 1, 10.2 PPF, Gas Tight Conn, seamless, 1.5m long, PSL3.	15	Ea.	MSA
46	Tubing pup joint, 4 1/2", L080 type 1, 18.9 PPF, Gas Tight Conn, seamless, 1.5m, PSL3.	5	Ea.	MSA
47	Liner Pup Joint, 5", L80 type 1, 18 PPF, premium flush connection, seamless, PSL3.	4	Ea.	MSA
48	7" Scraper	1	Ea.	OEOC
49	9-5/8" Scraper	1	Ea.	OEOC
50	5" DP,HWDP,8-1/2" DC,	as required		OEOC
51	Well Head Services	as required		Service Co.
52	Surface Well Test Services	as required		Service Co.
53	Completion Services	as required		Service Co.

54	Slick line Services	as required		Service Co.
55	Casing Running Services	as required		Service Co.
56	Wireline Logging	as required		Service Co.
57	Dry Test, DST, RTTS	as required		Service Co.
58	Cementing Services	as required		Service Co.
59	Stimulation (Coiled Tubing, Acidizing and N2) Services	as required		Service Co.
60	H2S Services	as required		Service Co.
61	Mud Logging Services	as required		Service Co.
62	Rig Services	as required		OEOC
63	Drilling & Completion Fluid Services	as required		Service Co.
	Note: Quantity & Spec to be conformed with contract's SOW. Note: In case of any shortage or discrepancy, the contract's SOW will be reference.			

Revision Corrections

Drilling Engineering							
۱	• در برنامه و جدول شکست زمانی لاگ SBT تحت فشار انجام نشود (نرمال اخذ گردد)		CM	اصلاح شد			
۲	• جهت راندن آستری ۵" ابتدا Scrap trip و سپس condition trip انجام گردد . در جدول شکست زمانی و برنامه اصلاح شود.		CM	اصلاح شد			
۳	• Positive test برای آستری های ۵" و ۷" در صورت گرفتن bumping نیز می بایست انجام شود.		CM	اصلاح شد			
۴	• جابجایی چاه با Cacl2 در ۲ مرحله آورده شده است. که مرحله اول اشتباه است.حذف گردد. • **Perform 5" Liner CIT. (80% of Burst press) with 2000 psi,for 15 min, (If not obs** • **bumping).CBU & Displace well with CaCl2, POOH to Surface.**		CM	اصلاح شد			
۵	• جهت برداشتن BOP و نصب تاج چاه دو عدد پلاگ		CM	اصلاح شد			

	درون رشته تکمیلی باید نصب گردد.						
۶	• قبل راندن آستری ۷" (در صورت داشتن پکر) می‌بایست Scrap trip قبل از condition trip انجام گردد.	۲۸	CM	انجام Scraper trip جهت راندن SN-Packer بعد از تست خشک لبه و مشخص شدن نشتی لبه آستری انجام خواهد شد.			
۷	• ردیف ۷۳، contingency #10 حذف گردد. • Note: If dry test is not ok, follow contingency plan #10.	۳۵	CM	تصحیح گردید			AP
۸	• عمق بالا آمدن لوله جهت انجام flush با صفحه ۳۵ ردیف ۶۷ یکسان سازی شود و عمق صحیح وارد گردد. • POOH w/ L.Hanger setting tool to 3000m, Flush string & CBU.	۲۹	CM	تصحیح گردید			AP
۹	در contingency#9 عمق لبه آستری به ۳۶۲۸ متری جهت انجام تست خشک لبه آستری ۵" اصلاح شود. Perform Scraper Trip, RIH with 7" RTTS & Dry test BHA, RIH to L.Lap @ +/- 3310m, Perform 5" Liner lap Dry Test • همچنین عمق لبه ۷" به ۲۱۶۵ اصلاح شود	31 , 37	CM	تصحیح گردید	در contingency#9 عمق لبه ۷" به ۲۱۶۵ اصلاح شود.		CM

	• عمق LDC آستری ۵" با صفحه ۳۵و 29 , در جدول یکسان سازی شود و به عمق صحیح اصلاح گردد.						
۱۰	• با توجه به تغییر برنامه و نهایی نشدن زمان بندی TBD، تا تائید نهایی زمانبندی عملیات از طرف کارفرما، زمان اعلام شده منوط به تایید کارفرما می‌باشد.		CM	تصحیح گردید	با توجه به تغییر برنامه و نهایی نشدن زمان بندی TBD، تا تائید نهایی زمانبندی عملیات از طرف کارفرما، زمان اعلام شده منوط به تایید کارفرما می‌باشد.		CM
۱۱	• در جدول شکست زمانی و متن صفحات ذکر شده برای آستری ۷" CIT با فشار ۳۰۰۰ PSI لحاظ گردد.	33-27	CM	تصحیح گردید			AP
۱۲	• در جدول شکست زمانی و متن صفحات ذکر شده FIT برای حفره جدید ۱/۸ ۶" با پمپ تراک لحاظ گردد.	33-27	CM	اضافه شد			AP
۱۳	• در جدول شکست زمانی صفحه ۲۸، عمق ناحیه نصب لاینر هنگر اصلاح و با صفحه ۳۴ یکسان سازی شود. • RIH w/ 7" Scraper clean out BHA and scrape around 3310 m for Liner Hanger	۲۸	CM	تصحیح گردید			AP

	setting area, POOH						
۱۴	• در contingency#9 عمق لبه آستری جهت انجام تست خشک لبه آستری ۵" اصلاح شود.	36	CM	تصحیح گردید			
۱۵	• در جدول لوله جداری، مشخصات آستری ۵" اضافه گردد.و CIT بر اساس آن مجددا محاسبه گردد.	۲۰	CM	با توجه به وزن گل در نظر گرفته شده و مشخصات آستری CIT می بایست با ۲۰۰۰ psiانجام گردد			AP
۱۶	• در جدول لوله جداری، اعماق آستری ۷" به ۲۱۶۵ تا ۳۷۲۸ اصلاح شود .	۲۰	CM	تصحیح گردید			AP
۱۷	• با توجه به متن قرارداد تامین سرویس پسماند الزامیست. • با توجه به اینکه در Program REV01 مقرر شده بود بعد از Side Track برنامه پسماند ارایه گردد،در Program REV02 اعمال نگردیده.		CM	برنامه مدیریت پسماند در دست تهیه می باشد و در اسرع وقت به صورت جداگانه به کارفرمای محترم ارسال خواهد شد			AN
۱۸	• هنگام تست دالیز، پلاگ و پرانگ از رشته تکمیلی خارج گردد.		CM	در صفحه ۸۲ اضافه گردید			AP
۱۹	• عمق نصب پکر در حداقل فاصله از بازه مشبک کاری قرار گیرد.		CM	در برنامه رانش تکمیل این موضوع لحاظ خواهد شد			AN

۲۰	• شماتیک بعد از کنارگذر به صورت افقی رسم و تکمیل گردد.			تصحیح گردید.			AP